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BOOK 7

Hydropower Development and Finance

Developing, structuring and financing hydropower projects from site to financial close

AUTHOR & IDEATION

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About this book

Book 7. Hydropower Development and Finance: Business Models, Project Development and Financial Structuring for Hydropower in Africa. First edition, version 0.1 (pre-publication).

The book belongs to the Africa Energy Finance collection, Business & Financial Models: a series of books, financial models, manuals, case studies and decision tools for energy businesses in African markets. Each book is built around one central question.

Book	Subject	Central question	Status
Book 1	Mini grids	Can this mini grid become a sustainable business?	Planned
Book 2	Solar home systems (PAYGo)	Can PAYGo solar become profitable and financeable?	First edition
Book 3	Clean cooking	Can clean cooking scale as a commercial business?	Planned
Book 4	Commercial and industrial solar with storage	Should the customer invest, sign a PPA or use an energy service company?	Planned
Book 5	Energy access fund	Can a fund mobilise capital and generate sustainable returns?	Planned
Book 6	Power utilities	Can the utility become financially sustainable?	Planned
Book 7	Hydropower: this book	Can this hydropower project be developed to financial close and financed on terms the developer, the lenders and the state can each accept?	First edition, v0.1

Each book is the centre of a set of products that share one method. For Book 7 they are:

Product	Name	Role
Book	Hydropower Development and Finance (this book)	Explains the method and why it works
Model	MODEL 7: Hydropower Development and Finance Model	Quantifies the method: annual, 40 period, integrated project, utility and public-finance model with a development module and 23 financial close gates
Manual	MANUAL 7: User and Methodology Manual	Teaches the reader to operate the model
Case	CASE 7: Kasiri River Hydro	Demonstrates the method on a fictional 60 MW run-of-river project
Training	Video course	Walks through the book and the model

The book is written to stand on its own. A reader who never opens the model loses the worked numbers, not the argument.

Preface

This book asks one question: can a given hydropower project be developed to financial close and financed on terms that the developer, the lenders and the state can each accept? Most hydro projects in Africa that never reach close do not fail on engineering. They fail because one of the three parties cannot accept the terms the other two need. The developer cannot earn enough to justify years of at-risk spending. The lenders cannot get comfortable with the river, the contractor or the buyer. Or the state is asked to carry obligations its budget cannot bear.

The book is written for the people who make those decisions. Developers and their finance directors building a development plan and a financing case. Credit officers and investment officers at development finance institutions and commercial banks sizing debt. Fund managers deciding whether to buy into a project before or after close. Transaction advisers who have to reconcile all of them. And staff in finance ministries and PPP units, who get one full chapter written for them, because a support package the ministry will not sign does not close.

Each chapter supports one decision. Chapter 1 sets out the business model and the reason why a hydro project has to be read as three investments, not one. Chapters 2 to 4 cover the resource and the commercial route: the river, the configuration and the path to a power purchase agreement. Chapters 5 to 8 deal with development: the stage gates, the development budget, the consents and the contract set. Chapters 9 to 11 turn to construction and operation. Chapters 12 to 15 are about financing: sizing the debt, assembling the lender group, the developer's returns and the public side. Chapters 16 and 17 bring the analysis together for the people who carry the risk, through stress testing and the financial close gates. Chapter 18 walks through the complete case.

How the book works with the model

The book has a companion workbook, MODEL 7, together with a user manual and a worked case on Kasiri River Hydro, a fictional 60 MW run-of-river project in the fictional Republic of Navaria, developed by the fictional Tamarind Hydro Ltd. The chapters refer to the workbook by sheet name, so that each concept can be tested on numbers. A reader who prefers to stay with the text can ignore those references. A reader who works through them will finish the book able to build, challenge and defend a hydro investment case.

All default inputs in the workbook, and all Kasiri figures, are illustrative. They were chosen to make the mechanics visible and to sit inside the ranges found in public sources, not to describe any real project. Where the book cites figures about real projects or about the sector, it says where they come from and how far they have been verified. Where public sources conflict, the book reports the conflict rather than choosing the more convenient number. Where no public figure exists, as for development premiums or stage-by-stage attrition in African hydro, the book says so and treats the value as a model assumption.

A second fictional project, Lumora Falls, a 400 MW scheme structured as a public-private partnership, appears as a contrast in Chapters 1, 15 and 16. It comes from the author's earlier policy paper on hydro and public liabilities, which is now a companion paper to this book.

Conventions

Currency amounts are in US dollars (USD) unless stated, and "USDm" means millions of US dollars. Unless stated otherwise, amounts in the model are nominal; capital costs quoted "real 2026" are before escalation. Energy is in GWh and capacity in MW. P50 and P90 denote the energy exceeded with 50 and 90 percent probability; "one-year" and "ten-year" P90 are distinguished in Chapter 2. Unit costs are labelled by scope (plant, funding base, total uses or system), because a USD/kW figure without a scope cannot be compared with anything. Negative amounts are shown with a minus sign. British spelling is used throughout.

Sources and evidence

The book draws on four kinds of material and labels each one. Institutional evidence comes from bodies such as the World Bank Group, the African Development Bank, IRENA and national regulators; project disclosures by development finance institutions are placed in the same class. Secondary reporting is trade press and law-firm guides that report those documents. A model assumption is an input of the companion workbook, chosen to make the mechanics visible. An illustrative case output is a figure produced by the workbook for Kasiri; it is never a benchmark for real projects.

Every external claim carries a source identifier in square brackets, such as [HY-08] or [DE:S1], which points to Annex L. The annex records, for each source, whether its full text was read, only its landing page, or only a search-engine summary. Claims resting on a summary are flagged as such and should be checked against the original before they are quoted.

Status of this edition

This is the first edition of Book 7, issued as version 0.1 for pre-publication review. It replaces, for developers and lenders, the author's earlier policy paper on structuring hydro without unsustainable public liabilities, whose material is condensed in Chapter 15. Three things must happen before release: the sources marked as summaries in Annex L must be read in full; the companion model must complete its test in Microsoft Excel (it has been tested in LibreOffice, adversarially reviewed and corrected); and practitioners must review the text.

Acknowledgements and disclaimer

The analysis and any errors are the author's. Nothing in this book is investment, legal, tax or accounting advice. Readers should take professional advice in the jurisdiction concerned before acting on any of it.

Introduction: how to read a hydro project

A hydro project can be read in a fixed order, and the order matters more than any single ratio. Each link in the chain below takes an input from the link before it and passes a result to the next. A weakness found early, in the flow record or the permitting sequence, travels down the chain and arrives, larger, at the debt sizing and the close.

Link	Question	What it passes on	Chapters
River and site	Is there a resource worth developing here?	Catchment, head, access, competing uses	2, 3
Hydrology evidence	How good is the flow record, and how uncertain is the energy?	Flow series, P50, one-year and ten-year P90	2
Configuration	What should be built?	Design flow, MW, energy by season, cost envelope	3
Offtake route	Who buys, through which route, on which tariff?	PPA route, tariff form, indexation, currency	4
Development plan	What must be done, in what order, at what cost and probability?	Stage plan, budget, stage-gate probabilities	5, 6
Consents	Which permits gate which others?	Consent sequence, critical path, E&S conditions	7
Contract set	Is the chain of contracts bankable and consistent?	Concession, PPA, grid, direct agreements, termination amounts	8
Construction	Who takes geology and delay, and at what price?	Contract structure, contingency, owner's share of overruns	9, 10
Operations	Who runs it, and what will it cost to keep it available?	O&M model, availability, reserves, insurance	11
Debt	How much senior debt, at what tenor, on which case?	Debt quantum, DSCR, LLCR, reserve accounts	12
Lender group	Which lenders, guarantees and insurances?	DFI club, cover, local-currency tranches	13
Developer returns	Is it investable for the developer, and at what price can it sell down?	Risk-weighted NPV, premium, IRR, sell-down value	14
Public obligations	What does the state commit, and can it carry it?	Payment capacity, support, contingent liabilities	15
Stress	What breaks first, and who holds it?	Break-even flow, overrun, distance to default	16
Financial close	Go, conditional go or stop, on what evidence?	Gate status, decision and conditions	17, 18

Eight questions that are not the same question

Hydro projects are often described as "feasible" or "bankable" as if those words meant one thing. The book separates eight questions, because a project can pass one and fail the next, and each is answered by a different document or test.

Question	Answered by	Typical trap
Is the site technically viable?	Feasibility study and geotechnical investigation (Chapters 2, 3, 9)	A feasibility study built on a short or synthetic flow record presented as a long one
Is the project developable?	Consent sequence, land and water rights, PPA route (Chapters 5, 7)	Spending on bankable feasibility before the PPA route and water rights are secure
Is it economic for the power system?	Least-cost plan, LCOE against alternatives (Chapters 3, 4)	Quoting plant cost when the system pays for plant and line
Is it investable for the developer?	Risk-weighted development NPV, premium, sell-down value (Chapters 6, 14)	Reading the equity IRR at close as the developer's return, ignoring at-risk spend and the chance of failure
Is it bankable?	Lender case, DSCR, LLCR, security package (Chapters 12, 13)	A project protected by backstops, so that lenders see none of the risk
Is it buildable on budget?	Contract structure, contingency, reference class (Chapters 9, 10)	A "fixed-price" EPC contract with ground conditions excluded
Is it affordable for the buyer and the state?	Payment capacity, support package, contingent liabilities (Chapter 15)	A bankable project whose arrangements move the risk onto the Treasury
Is it ready to close?	Evidence-based financial close gates (Chapter 17)	Treating a signed term sheet as a closed deal

The rest of the book follows the chain in order.

Chapter 1. The hydro project: three investments in one asset

This chapter supports one decision, which comes before any other in the book: which kind of capital should fund which phase of a hydro project, and therefore which numbers each party should read first. A project funded as if it were one investment will be priced as one investment, and the price will be wrong for at least two of its three phases. The chapter sets out the three investments that sit inside one hydro project, shows where value and cash are created and where they diverge, describes who develops hydro in Africa and how projects fail, and explains how the rest of the book and the companion workbook are organised.

Place in the analytical chain: the frame for the whole chain, the three investments inside one project.

1.1 Three investments in one asset

Seen from a distance, a hydro project is a single asset: a weir or dam, a waterway, a powerhouse and a line to the grid, owned by one special-purpose company that sells electricity under one power purchase agreement. Inside, it is three investments made in sequence, each with its own risks, its own investors and its own cost of capital.

The first is a development option. Before anything is built, a developer spends money on studies, flow gauging, permits, land, negotiations and advisers, for years, with no assurance that the project will ever reach financial close. If it does, the developer is reimbursed and usually earns a premium. If it does not, the money is lost. This is venture capital in everything but name, and it should be priced like venture capital.

The second is a construction contract. From financial close to commercial operation, the project's owners and lenders are exposed to geology, contractor performance, delay and the interface with the transmission line. The money at stake is far larger than in development, and the risk is different: it is the risk that the asset costs more or takes longer than planned. In hydro, more than in any other power technology, that risk sits underground.

The third is an operating annuity. Once the plant runs, the project becomes a long-dated claim on two things: the river and the buyer. Revenue depends on how much water flows and on whether the utility pays. Lenders and long-term holders of equity price this phase, and they price it at a much lower cost of capital than development or construction, provided the river and the buyer have been understood.

Area	Development option	Construction contract	Operating annuity
Period	Site to financial close, often five years or more	Close to commercial operation	Commercial operation to end of concession
Main risks	Permits, PPA, land, hydrology evidence, failure to close	Geology, contractor performance, delay, interface with the line	Hydrology, offtaker payment, currency, O&M, regulation
Who funds it	Developer equity, co-developers, project preparation facilities	Equity, senior debt, contingency, standby equity	Senior debt service, distributions, refinancing
How value shows	Step-ups at milestones; premium at close	Completion; liquidated damages; contingency release	DSCR, distributions, refinancing gains
Typical failure	Spending past a gate the project should have failed	Overrun beyond contingency; contractor default	Offtaker arrears; a run of dry years; a currency step

Many of the costly mistakes in hydro come from funding one of these investments with capital meant for another. Development spending financed by a utility's operating budget is stopped when the budget is cut. Construction overruns financed by lenders who sized their debt on the operating annuity trigger a default. Operating cash flows priced at development-stage returns produce tariffs that the buyer cannot pay.

1.2 How value and cash are created

Value in a hydro project is created at three moments, and cash arrives at none of them.

The first is financial close. On the day the contracts are signed and the debt becomes available, a set of permits, studies and agreements that was worth almost nothing to an outsider the day before becomes a financeable project. The developer captures part of that value through the reimbursement of its development costs and a development premium, and part through its share of the equity.

The second is commercial operation. Once the plant is built and running, construction risk has gone. Investors who would demand a construction-stage return will now accept an operating-asset return, and the same future cash flows are worth more. That step-up is what allows a developer to sell part of its stake after commercial operation and recycle the proceeds into its next project.

The third is the long tail of operation. Revenue arrives monthly, debt service is paid twice a year, and distributions start only when the lenders' tests are met.

The Kasiri case shows the shape. Tamarind Hydro spends about USD 9.0 million, in 2026 dollars, over 6 years of development. In the model, the chance that a project at the reconnaissance stage reaches close is about 15%, a model assumption informed by the evidence in Chapter 5. If Kasiri closes, total uses of funds are about USD 215 million, of which senior debt provides about USD 145 million. On the success path, the developer's internal rate of return over the whole period, from the first study to the last distribution, is 16.0%, including a sale of half its stake at commercial operation. Weighted by the chance of failure, the same developer's expected net present value at a 25 percent discount rate is USD -0.95 million.

Those two numbers describe the same project. The first answers the question "what does the developer earn if it succeeds?" The second answers "should a developer start?" Chapter 14 is about the gap between them.

1.3 Who develops hydro in Africa

Three kinds of developer take hydro projects to close in Sub-Saharan Africa.

State utilities and governments develop most large schemes, often with export-credit or bilateral finance and an engineering, procurement and construction contractor who also arranges the loan. Julius Nyerere in Tanzania was funded from the state budget [A2:S30]. Isimba and Karuma in Uganda were 85 percent financed by China Exim [A2:S34; A2:S40].

Private developers, often regional companies backed by development finance institutions, develop most small and medium plants. The World Bank's 2024 review of private participation in large hydro found that the private share of hydro capacity falls sharply with plant size [HY-06]. Uganda's GET FiT programme supported a portfolio of small private hydro plants, with about EUR 94 million of public money mobilising about EUR 450 million of private investment according to KfW [DE:S35].

Development finance institutions increasingly co-develop. IFC InfraVentures typically funds a minority share of development costs and an average ticket of about USD 4 million per project [DE:S2]. The African Development Bank's Sustainable Energy Fund for Africa has provided project preparation grants of around USD 1 million to small hydro [DE:S17].

This book is written mainly for the second and third groups, and for the lenders and ministries who deal with them. Its main case is a 60 MW private project because that is the scale at which a private developer is the protagonist.

1.4 How hydro projects fail

Hydro projects fail in three patterns, one for each investment.

Development stalls. Ngonye Falls in Zambia has been in development for more than a decade without reaching financial close [DE:S3]. Ruzizi III, a regional public-private partnership, has been stalled since about 2009 [A2:S20]. Most of the money spent on such projects is never recovered.

Construction overruns. The reference class for large dams shows a median real cost overrun of 27 percent and a mean of 96 percent [HY-08]. African examples include a 27 percent cost increase at Bui [A1:S38], a 47 percent increase in the main civil works contract at Rusumo [A2:S16] and a commissioning date several years late at Karuma [A2:S40]. The reference class covers large dams, not medium run-of-river plants, so it is a warning rather than a calibrated forecast for a project like Kasiri.

The operating annuity breaks. A run of dry years cuts revenue. At Kariba in 2024 the water allocation for generation was cut by about half, according to news reporting [CL-04]. A buyer that cannot pay leaves arrears: Ghana's ECG owed the Bui Power Authority about USD 612 million in March 2023 [A1:S40]. A currency step multiplies the local-currency cost of a dollar tariff overnight.

None of these failures is visible first in the project's income statement. All of them are visible first in the development budget, the flow record, the contract structure and the buyer's cash flow.

1.5 A contrast: Lumora Falls

The author's earlier policy paper used a fictional 400 MW public-private partnership, Lumora Falls, as its case. Lumora's tariff put about 71 percent of revenue into a capacity payment, which the utility owes whether or not the river flows. As a result, a three-year drought barely moved Lumora's equity return, and the paper's main finding was about the state, not the developer: the project's own debt cover stayed robust while the arrangements that made it so moved risk onto the utility and the Treasury.

Kasiri is the opposite case. Its tariff is paid only for energy delivered, as under East African feed-in tariffs, so the river is the developer's problem. Its size means that the state's exposure is small relative to the budget. And its developer has to decide, at each stage, whether to keep spending. The two cases together show why the same method gives different answers for different projects, and why the reader should never transfer a conclusion from one to the other without checking the tariff structure first.

1.6 How the book and the workbook are organised

The book has 18 chapters, built around the central question. Chapters 2 to 4 cover the resource and the commercial route. Chapters 5 to 8 cover development: the stage gates, the budget, the consents and the contracts. Chapters 9 to 11 cover construction and operation. Chapters 12 to 15 cover financing, from the lender case to the public side. Chapters 16 and 17 cover stress testing and financial close, and Chapter 18 walks through the Kasiri case from site to close.

The companion workbook, MODEL 7, is an annual, 40 period model with 38 sheets and about 11,000 formulas, no macros and no circular references. It links hydrology, transmission, demand, the utility's cash flow, the PPA, project finance, five financing structures, government support, contingent liabilities and a fiscal screen. A development module covers the years before close, a contracting module compares construction contract structures, and a readiness sheet applies 23 financial close gates. Its 14 integrity checks confirm internal consistency, not the realism of the assumptions. The model was reviewed adversarially before this edition; the review found and fixed defects that the model's own checks had missed, and Annex K records what was changed.

The workbook never rates a project. Its readiness sheet counts each gate only on evidence and applies a fixed rule: STOP when a critical gate is not met or its evidence is missing, CONDITIONAL GO when every critical gate is met, GO only when all gates are met. In its default state, Kasiri meets 6 of the 23 gates and reads "STOP: a critical gate is not met". That is a finding about the state of the project's evidence, not a judgement on the project.

Points for the investment and credit committees

1. Ask the sponsor to present the project as three investments, with the development budget, the construction budget and the operating case shown separately, each with its own funding and its own target return.
2. Read the developer's return twice: on the success path and weighted by the probability of failure. Treat any presentation that shows only the first as incomplete.

3. Identify who carries the river. If the tariff pays for energy, the flow record is the core of the credit; if it pays for capacity, the buyer's balance sheet is.
4. Check that construction risk is funded by capital that can absorb it: contingency, standby equity or an overrun facility, not senior debt sized on the operating case.
5. Test the buyer's ability to pay before negotiating the tariff, and decide openly who funds any shortfall.
6. Do not transfer conclusions from a large public-private scheme to a mid-size private project, or the reverse, without checking the tariff structure and the size of the state's exposure.

Working with the model

Open *01_CONTROL_PANEL* first: it sets the case, the generation case for cash flows and for debt sizing, the financing structure and the stress toggles. Then read *32_DASHBOARD*, which shows the project, financial, developer, power system, utility and public-finance results side by side. *01A_DEVELOPMENT* holds the development budget, the stage probabilities and the developer's returns; *05A_CONTRACTING* compares construction contract structures; *30A_CLOSE_READINESS* applies the 23 gates. *33_CHECKS* should read ALL OK before any output is relied on.

Chapter 2. River, flow record and hydrology risk

This chapter supports a decision that is taken twice: is the hydrological evidence good enough to spend feasibility money, and later, is it good enough to size debt? The two thresholds are different. A developer can justify a feasibility study on a short record and a regional correlation; a lender cannot justify a sixteen-year loan on the same evidence. The chapter explains how flow becomes energy, how uncertainty in energy is measured, why a one-year P90 and a ten-year P90 answer different questions, and why the tariff structure decides who carries the river.

Place in the analytical chain: hydrology evidence, the first link that every later number depends on.

2.1 Four kinds of capacity

Financing documents are full of capacity numbers, and they do not mean the same thing. Installed capacity is the nameplate rating of the generating units. Available capacity is installed capacity multiplied by the share of time the units can run, after planned and forced outages. Expected generation is the energy the river allows on average, which depends on flow, head and how often the flow exceeds what the turbines can use. Contracted energy is whatever the PPA refers to, and it may differ both from what is delivered, because of curtailment, and from what is paid for, because of deemed-energy clauses.

The gap between nameplate and energy can be large. Belo Monte in Brazil has 11,233 MW installed against 4,571 MW of assured energy [INT:S8]. For a run-of-river plant like Kasiri, the gap is set by the shape of the river's seasons: in the dry months the turbines run well below their rating.

2.2 The flow record

Everything in this book that depends on energy depends on the flow record. The IFC's guide for hydro developers sets a minimum of about 15 years of hydrological data for lenders [DE:S1]. Few African sites have that much measured data at the intake. Most developers start a gauging programme at pre-feasibility and extend the short on-site record by correlation with a longer regional gauge.

That practice is sound if it is done well and described honestly. The questions a lender's adviser will ask are concrete. How many years were measured on site, and how many were synthesised? How good is the correlation, and over which flow range was it tested? Were the rating curves checked at high flows, where most of the uncertainty sits? Are there gaps, and how were they filled? Has upstream land use, abstraction or a new dam changed the river since the regional record began?

In the model, the record is described by two inputs on *03_HYDROLOGY*: the number of reliable years and the maturity of the hydrology study, from desk study to independent review. Kasiri has 12 years, a mix of on-site gauging since pre-feasibility and regional correlation, and a feasibility-level study that has not yet been independently reviewed. Gate 1 of the financial close readiness sheet therefore reads NOT MET. That is the normal state of a project at Kasiri's stage, and it tells the developer where to spend next.

2.3 From flow to energy

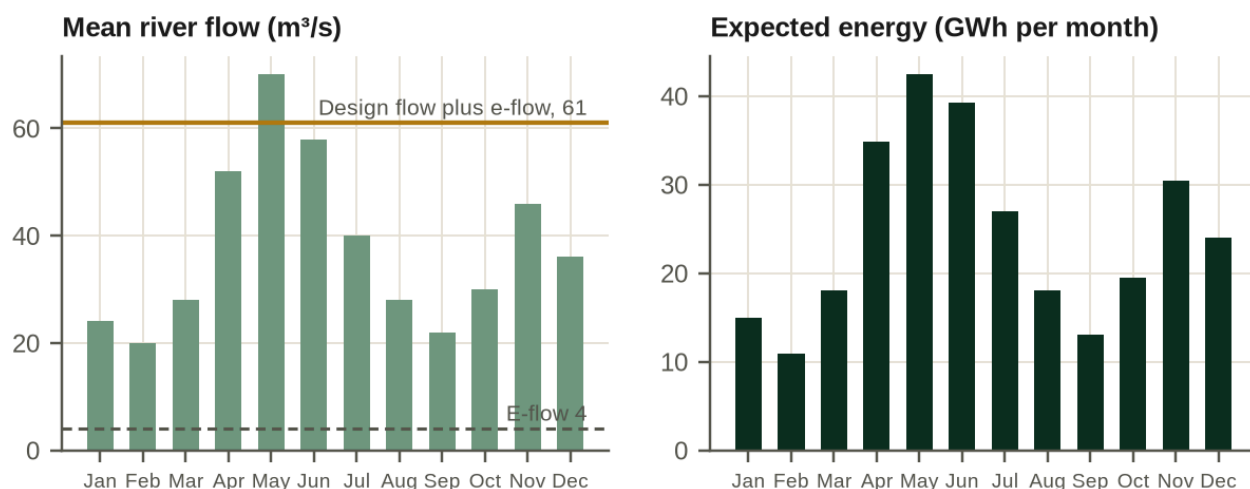
For each month, the model computes usable flow, average power and energy:

$$Q_{\text{usable}} = \min(\max(Q - Q_{\text{env}}, 0), Q_{\text{design}})$$

$$P = \min(\rho \cdot g \cdot Q_{\text{usable}} \cdot H \cdot \eta_{\text{turbine}} \cdot \eta_{\text{generator}} / 10^6, P_{\text{installed}})$$

$$E = P \times \text{hours} \times \text{availability} / 1000$$

Q is mean river flow in m^3/s , Q_{env} the environmental flow release, Q_{design} the rated turbine flow, H the net head in metres, η the efficiencies, P in MW and E in GWh. For Kasiri, mean monthly flows range from 20 to 70 m^3/s , the design flow is 57 m^3/s , the environmental release 4 m^3/s and the net head 120 m. The long-term mean energy, the P50, is about 292 GWh a year, a capacity factor of 55.6%. IRENA reports a weighted average capacity factor of 55 percent for small hydro commissioned in Africa between 2018 and 2024, within a 5th to 95th percentile band of 50 to 66 percent [HY-01].



The equations hide a bias that matters for finance. Energy is a concave function of flow: it rises with flow until the turbines are full and then stops. Applying the function to mean monthly flows overstates mean energy, because the water spilled in wet years cannot make up for the energy lost in dry ones. For a bankable estimate, energy should be computed month by month over the whole record and then averaged. The model accepts a simulated series when one exists and flags the bias when it does not.

2.4 P50, one-year P90 and ten-year P90

Lenders size debt on a downside generation case. The usual reference is a P90: the energy exceeded with 90 percent probability. Without a full simulated series, the model uses the coefficient of variation (CV) of annual energy:

$$P90_{\text{one-year}} = P50 \times (1 - 1.2816 \times CV)$$

With a CV of 15 percent, Kasiri's one-year P90 is about 236 GWh.

A one-year P90 describes a single bad year. A sixteen-year loan is serviced by many years, and the average of many years is much less uncertain than any one of them, unless dry years cluster. The model therefore also computes the P90 of the ten-year average, allowing for persistence through a year-to-year correlation coefficient ρ :

$$P90_{\text{ten-year}} = P50 \times (1 - 1.2816 \times CV \times \sqrt{((1 + \rho) / (1 - \rho) / 10)})$$

With $\rho = 0.3$, Kasiri's ten-year P90 is about 268 GWh, much closer to P50 than the one-year figure.

The choice between the two is not technical. It is a negotiation about who carries persistence risk. Sizing debt on the one-year P90 for every year of the loan is conservative for the average and says nothing about runs of dry years; sizing on the ten-year P90 is more generous but needs a separate test for a multi-year drought. The model's default sizes on the ten-year P90 and then tests a three-year drought explicitly, with the debt service reserve available. Chapter 12 shows the effect on debt capacity.

2.5 Who carries the river

The tariff structure decides who carries hydrology risk. Under a two-part tariff, a capacity charge pays the fixed costs whether or not the river flows, and the buyer carries most of the hydrology risk. Under an energy-only tariff, as in East African feed-in tariffs, the project is paid only for what it delivers, and the river is the project's problem [DE:S8; DE:S42].

Kasiri has an energy-only tariff. With the financing package fixed at its base-case terms, a three-year drought at 55 percent of normal flow from the fourth operating year reduces the minimum DSCR from 1.53x to 0.69x and the equity return from 13.8% to 11.0%. Running the whole cash flow on the one-year P90 instead of P50 gives a minimum DSCR of 1.16x and an equity return of 8.1%. The contrast with Lumora, where a similar drought barely moved the equity return, is the tariff, not the river.

The drought factor deserves a word. In 2024 the water allocation for generation at Kariba was cut from 30 to 16 billion cubic metres, according to news reporting [CL-04]. The model's default of 55 percent of normal flow for three years is set close to that event. A single-year statistic would never produce it.

2.6 Climate

Peer-reviewed work warns that planned hydro expansion in eastern and southern Africa concentrates capacity in basins with correlated rainfall, which raises the risk of concurrent supply disruption across interconnected systems [CL-03]. For a single project, the practical consequence is that the historical record may not describe the next thirty years. The IHA's climate resilience guide sets out how to test a design against climate scenarios [CL-01]. The model offers a flow trend per decade as a stress, labelled illustrative until a basin study supports a number. With a decline of 3 percent per decade, Kasiri's equity return falls from 13.8% to 12.8%.

2.7 What the Kasiri case shows

Kasiri's hydrology is adequate for a developer to keep spending and inadequate for a lender to commit. The record is shorter than 15 years and has not been independently reviewed. Because the tariff pays only for energy, the project's credit rests on that record. The next development money should go into extending the record, checking the rating curves and commissioning the independent review, before the bankable feasibility study is finalised.

Points for the investment and credit committees

1. Ask how many years of the flow record were measured at or near the intake, and how many were synthesised by correlation. Treat the two differently.
2. Require energy to be computed month by month over the whole record, not from mean monthly flows.
3. Ask for both the one-year and the ten-year P90, and agree explicitly which one sizes the debt and how runs of dry years are tested.
4. Identify who carries hydrology risk under the tariff, and size reserves accordingly.
5. Test at least one multi-year drought set close to the worst sequence in the region's record.
6. Treat a climate trend as a stress, not a base case, unless a basin-specific study supports it.

Working with the model

03_HYDROLOGY holds the monthly flows, the plant parameters, the P-values and the record inputs. *04_GENERATION* shows installed capacity, available capacity, generation, evacuable energy, delivered energy and contracted energy year by year. On *01_CONTROL_PANEL*, switch the generation case for cash flows between P50, P75 and P90, and the lender sizing case between one-year and ten-year P90. Switch on the drought and climate toggles and read the minimum DSCR on *20_CASH_FLOW*. Gate 1 on *30A_CLOSE_READINESS* reads the record inputs automatically.

Chapter 3. Configuration, energy and the cost envelope

This chapter supports the decision on what to build: the design flow, the installed capacity and the scheme type, and therefore the capital cost envelope that every later chapter will try to finance. Configuration is usually treated as an engineering question. It is also a financing question, because each extra megawatt of installed capacity adds cost in every month but energy only in the wet ones.

Place in the analytical chain: configuration, between the hydrology evidence and the offtake route.

3.1 Sizing the design flow

The design flow is the flow at which the turbines reach their rating. A larger design flow captures more of the wet-season water and raises annual energy, but each additional cubic metre per second is used for fewer months a year. The marginal megawatt-hour therefore costs more than the average one.

The standard tool is the flow duration curve: the share of time each flow is exceeded. A design flow exceeded 30 percent of the time captures most of the high-flow water but runs at part load for most of the year; one exceeded 60 percent of the time runs near full load more often but spills more water. The right choice depends on the tariff. Under an energy-only tariff with no seasonal price, energy is worth the same in every month and the developer maximises net present value by adding capacity until the cost of the last megawatt equals the value of the energy it adds. Under a tariff that pays more for dry-season energy, or a capacity charge, the optimum moves.

Kasiri's design flow is 57 m³/s. Only in May, the wettest month, does the river exceed the design flow plus the environmental release, and the excess is spilled; in April and June the plant runs close to its rating. In the four driest months it runs at between about a quarter and two fifths of its rating.

3.2 Run-of-river, pondage and storage

Kasiri is a run-of-river scheme with a small daily pond. It can shift some energy within a day, from night into the evening peak, but not between seasons. A storage scheme can move water from the wet season into the dry season, which raises firm energy and its value to the system, at the cost of a larger dam, a larger reservoir, more resettlement and more geological risk.

For a private developer, storage changes the financing problem in three ways. It raises capital cost per kilowatt. It raises the environmental and social stakes, and with them the time and cost of consents. And it makes the project more valuable to the system, which argues for a two-part tariff that pays for the firm capacity storage provides. None of these is a reason not to build storage. All of them are reasons to price it explicitly.

3.3 Capacity factor and firm energy

IRENA reports capacity factors for newly commissioned African large hydro of 33, 48 and 81 percent at the 5th percentile, weighted average and 95th percentile, and 50, 55 and 66 percent for small hydro [HY-01]. Kasiri's 55.6% sits at the small-hydro average.

Annual energy is not the whole story. The utility's planner cares about what the plant delivers in the driest month, because that is when the system is short. Kasiri's dry-season firm output, the lowest monthly average after availability, is about 16 MW against an installed 60 MW. A planner comparing Kasiri with a gas plant or a solar plant with storage would credit it with that firm figure, not the nameplate.

3.4 The cost envelope

IRENA's cost database gives a weighted average installed cost for large hydro (above 10 MW) in Africa of USD 2,330 per kW for projects commissioned in 2010 to 2017 and USD 2,515 per kW for 2018 to 2024, in 2024 dollars, and about USD 4,084 per kW for small hydro [HY-01]. Project-level costs vary much more widely. Uganda's GET FiT-era small plants

range from about USD 1,700 per kW at Nyamagasani 1 to about USD 6,200 per kW at Kikagati (derived from reported costs) [DE:S39b; DE:S53]. The IFC's guide gives a global small-hydro range of USD 1,300 to 8,000 per kW [DE:S1].

Kasiri's base plant cost is about USD 2,614 per kW in real 2026 terms, including development costs, the contractor's risk premium and a 10 percent contingency. That is a model assumption chosen inside the envelope, not a benchmark. The contingency follows the IFC's median of 9.8 percent of total plant cost, within a range of 5.4 to 12.6 percent [DE:S1].

Item	USDm, real 2026	Share
Civil works (weir, intake, tunnel, penstock, powerhouse)	68	Largest and most uncertain item
Hydro-mechanical and electro-mechanical equipment	42	Priced in hard currency
Environmental and social, engineering, owner's costs	17	Includes resettlement and supervision
Development costs reimbursed at close	9.0	From the development budget, Chapter 6
Contractor risk premium and contingency	balance	Chapter 9 and 10
Total plant cost	157	

A cost figure is only comparable when its scope is stated. The model reports four measures separately: plant cost, the funding base (adding any project-built transmission and the development premium), total uses (adding interest during construction, fees and the initial debt service reserve), and system cost (adding publicly funded transmission). For Kasiri these are about USD 157 million real, USD 187 million and USD 215 million nominal.

3.5 Plant cost and system cost

The model computes two levelised costs of electricity at a 10 percent nominal discount rate: the plant LCOE, about USD 106 per MWh, and the system LCOE, which adds any publicly funded transmission and the energy lost on the line, about USD 108 per MWh. Because Kasiri's developer builds its own 35 km interconnector, the two are close. For a project whose line the state builds, the difference can be large and sits on the public side of the ledger.

Points for the investment and credit committees

1. Ask to see the flow duration curve and the reason for the chosen design flow, expressed in money: the cost of the last megawatt against the value of the energy it adds.
2. Confirm whether the tariff values firm or dry-season energy, and whether the configuration responds to that.
3. Read unit cost only with its scope, and compare like with like.
4. Check the contingency against the uncertainty in the civil works, not as a single percentage.
5. Ask for the firm dry-season output, not only the capacity factor.

Working with the model

03_HYDROLOGY holds the plant parameters and shows monthly usable flow, power and spill. *05_PLANT_CAPEX* holds the cost breakdown and the benchmark comparison; development costs come from *01A_DEVELOPMENT* and the contractor's premium from *05A_CONTRACTING*. *17_PROJECT_FINANCE* reports the LCOE measures, and *31_CASE_STUDY* compares Kasiri's unit cost with public benchmark projects.

Chapter 4. The route to a PPA: procurement, tariff and the buyer

This chapter supports the decision on which offtake route to pursue and which tariff structure to propose. The route decides how long development will take, which lenders can participate and how much negotiating power the developer has. The tariff structure decides who carries the risk, the currency and the buyer's credit.

Place in the analytical chain: the offtake route, which passes the tariff, its currency and its risk allocation to every later link.

4.1 Four routes to a PPA

There are four routes, and most countries use more than one.

A standardised tariff, often called a renewable energy feed-in tariff, offers a published price and a standard PPA to plants below a size threshold. It shortens negotiation and lowers development cost, but the price is fixed by the regulator, not by the project's costs. Uganda's REFIT 5.0 sets small-hydro tariffs of about USD 0.0751 per kWh for 10 to 20 MW and USD 0.0792 per kWh for 0.5 to 5 MW, with 20-year PPAs, deemed energy and 60-day payment, according to a 2024 law-firm guide [DE:S8]. Kenya's 2012 feed-in tariff policy offered USD 0.105 per kWh at 0.5 MW, falling to USD 0.0825 at 10 to 20 MW, in US dollars for 20 years [DE:S42]. Tanzania's standardised small power PPAs cover plants up to 10 MW [DE:S44].

A competitive tender sets the price by competition. It is the route lenders and development finance institutions prefer, because it answers the question of value for money before they have to. Nyagak III in Uganda was awarded by tender [A2:S45].

Direct negotiation, including unsolicited proposals, is the most common route for plants above the standard-tariff threshold. It allows a tariff set on the project's costs, but it exposes the developer and the buyer to the charge that the price was not tested, and some lenders will not participate without an independent review of the tariff.

Bilateral sales to a large industrial buyer, such as a mine, bypass the utility for part of the output. They can improve the credit, if the buyer is strong, and they need a wheeling arrangement with the grid owner.

Kasiri, at 60 MW, is above the standard-tariff thresholds in the region. Tamarind is negotiating directly with the national utility, with the regulator's approval of the tariff as a condition of the PPA.

4.2 Procurement as a lender filter

The route matters to lenders for a reason that has little to do with price. At Gibe III in Ethiopia, the main civil works contract was awarded without competitive tender, which was incompatible with World Bank procurement policy; the African Development Bank and the European Investment Bank withdrew and the electro-mechanical package was financed by a Chinese commercial bank [A1:S53; A1:S54]. The first attempt at Bujagali collapsed after export credit agencies withdrew and corruption investigations followed [A2:S48]. At Batoka Gorge, Zambia withdrew from the 2019 arrangement in 2023, citing improper procurement [A2:S57].

A developer pursuing direct negotiation should therefore plan, from the start, how it will demonstrate value for money: an independent tariff review, a published cost benchmark or an open procurement of the construction contract.

4.3 Tariff structure

The main choice is between an energy-only tariff and a two-part tariff.

An energy-only tariff pays a price per MWh delivered. The project carries hydrology risk, and its lenders size debt on a downside energy case. It is simple, it is what feed-in tariff regimes in the region use, and it gives the project an incentive to maximise output.

A two-part tariff pays a capacity charge per kW per month, as long as the plant is available, and an energy charge per MWh. The buyer carries most of the hydrology risk. Lumora's two-part tariff put about 71 percent of revenue on the capacity side, which is why its equity return barely moved in a drought. The protection is not free. In a drought the buyer pays capacity charges for a plant that produces less and buys replacement energy elsewhere.

Kasiri's PPA is energy-only at USD 112 per MWh in 2026 dollars, half indexed to US inflation, for an assumed term of 25 years. That is well above the standard tariffs quoted above, for two reasons: Kasiri's unit cost is higher than the plants those tariffs were set for, and it is not eligible for a standard tariff. Chapter 12 shows how close the tariff is to the level the project needs, and Chapter 14 shows how the developer's return depends on it. A tariff of 116 USD per MWh in 2026 dollars would bring the private equity return to the 15 percent target.

4.4 Indexation and currency

A dollar tariff moves currency risk from the project to a buyer whose revenue is in local currency. The model lets part of the tariff be denominated in local currency and indexed to local inflation. Tanzania's standardised PPAs set tariffs in dollars but pay them in shillings [DE:S44]. Kenya's feed-in tariffs were set in dollars [DE:S42].

For the project, indexation matters as much as currency. A tariff that is fixed in nominal terms loses value every year; a tariff fully indexed to US inflation raises the buyer's cost in local terms every time the currency weakens. Kasiri's 50 percent indexation is a compromise that the buyer's payment capacity, tested in Chapter 15, can absorb in the base case.

4.5 Volume, payment and support

Three further terms decide how much of the tariff the project actually collects.

Deemed energy pays for energy the plant could have delivered but the grid or the buyer could not take. Uganda's REFIT includes it [DE:S8]. Without it, a transmission delay or a grid constraint becomes the project's loss.

Payment security covers the risk that the buyer pays late. A letter of credit or an escrow sized to a number of months of billing is the minimum lenders expect. The model treats six months as ready and three as conditional; Kasiri has three.

Results-based support can bridge the gap between the tariff a project needs and the tariff a buyer can pay. Uganda's GET FIT programme paid a premium of about USD 0.014 per kWh for small hydro, calculated on 20 years of output but paid within five years, half at commercial operation [DE:S9]. Programmes of that kind change the developer's cash flow profile more than its long-run returns, because they arrive early, when equity is most expensive.

4.6 What the Kasiri case shows

Kasiri's route is direct negotiation with a utility whose payment capacity, in the base case, is several times the project's bill. The tariff is energy-only and close to, but below, the level the project needs for its sponsors' target return. The developer's negotiating priorities are clear from the model: a modest increase in the tariff or its indexation, six months of payment security, deemed energy for grid curtailment, and an independent tariff review to answer the procurement question before lenders ask it.

Points for the investment and credit committees

1. Ask which route the project is on and why, and how value for money will be demonstrated if the route is direct negotiation.
2. Read the tariff structure as a risk allocation: who carries hydrology, currency and curtailment.
3. Compare the tariff with the level the project needs for its target return under the lender case, and treat a gap as a financing problem to solve, not a detail.
4. Require deemed energy and payment security in the PPA before relying on it.
5. Check whether a results-based premium or grant is available, and model its timing, not just its value.

Working with the model

14_PPA holds the tariff, indexation, currency share, deemed-energy and termination terms. *15_TARIFF* shows the tariff path and the utility's collected revenue per MWh. *16_REVENUE* shows billed and cash revenue and the routing of any shortfall. The required tariff for each financing structure is reported in the full-engine snapshot on *17A_STRUCTURES*.

Chapter 5. The development process and its stage gates

This chapter supports a decision taken at the end of every development stage: continue, change course or stop. Development money is the most expensive money in a hydro project, and the most common way to lose it is to keep spending after the project should have failed a gate. The chapter sets out the stages, what each must prove, how long they take, how often projects fail at each, and how to turn those facts into a stage-gate discipline.

Place in the analytical chain: the development plan, the link between the offtake route and the money spent to reach close.

5.1 Six stages from site to close

The model divides development into six stages. The names vary between developers and between guides, but the sequence does not.

Stage	What it must prove	Kasiri duration (months)	Kasiri budget (USDm, real)	Kasiri P(success to next stage)
1. Site identification and reconnaissance	That a site with useful head, flow and access exists, with no obvious fatal flaw	6	0.10	60%
2. Pre-feasibility and start of flow gauging	That a plant of a given size could produce energy at a cost the market might pay	12	0.60	60%
3. Feasibility, ESIA, geotechnical and grid studies	That the chosen configuration is technically sound, environmentally and socially acceptable, and connectable	18	4.50	70%
4. Licences, water rights, land and permits	That the project has the legal right to build and operate	12	0.80	85%
5. PPA, implementation agreement and tariff	That someone will buy the energy, on terms that can be financed	12	0.80	80%
6. Financing and close	That lenders and equity will commit on those terms	12	2.20	85%

The durations, budgets and probabilities are model assumptions for Kasiri, informed by the evidence in this chapter and the next. They are not statistics. In practice the stages overlap: the ESIA runs alongside the feasibility study, and PPA negotiations often start before permits are complete. The model treats them as sequential to keep the arithmetic transparent; a developer who can run them in parallel shortens the timeline and changes the value of the project, a point Chapter 14 returns to.

5.2 How long it takes

The IFC's guide for hydro developers gives indicative durations for small hydro: one to six months for site identification, about a month for pre-feasibility (three to six for larger plants), and two to six months for the feasibility study (nine to eighteen for medium and large plants), with procurement of the construction contract taking up to eighteen months for larger projects [DE:S1]. Those durations cover the studies themselves, not the waiting between them. An IFC InfraVentures presentation put the total time from development to commercial operation for hydro at eight to ten years [DE:S2]. Ngonye Falls in Zambia has been in development for more than a decade without reaching financial close [DE:S3], and Nyagak III in Uganda took about ten years from award to commissioning [DE:S4].

Kasiri's six stages take 6 years in the model, followed by three years of construction. That places it in the middle of the observed range.

5.3 How often projects fail

There is no public, stage-by-stage attrition data for African hydro. The closest evidence is general. Brookings, citing McKinsey, reported that about 80 percent of African infrastructure projects fail at the feasibility and business-plan stage, and that only about 10 percent reach financial close [DE:S23]. Castalia, reported in Engineering News, put the share of African renewable energy projects that reach financial close below 30 percent [DE:S18].

Kasiri's stage probabilities multiply to a chance of about 15% that a project at reconnaissance reaches close. That sits between the two published figures. The probabilities are a model input, and the right way to use them is to ask what they imply, not to argue about the third decimal. At 15%, a developer who wants one closed project should expect to start about 7.

5.4 What a gate is

A stage gate is a decision point with three possible outcomes: proceed, proceed with conditions, or stop. It is useful only if three things are fixed before the stage starts: the questions the stage must answer, the evidence that counts as an answer, and the person who decides. Without them, the gate becomes a formality and spending continues by default.

Gate	Questions to answer	Evidence that counts
End of reconnaissance	Is there head, flow and access? Is anything obviously fatal (protected area, international river, unresolved land claim)?	Site visit, maps, regional flow data, preliminary land and environmental screening
End of pre-feasibility	Is the energy cost competitive with the likely tariff? Is the PPA route plausible?	Preliminary layout and cost, energy estimate from correlated flows, discussion with the buyer and regulator
End of feasibility	Is the design sound? Is the ESIA approvable? Is the grid connection feasible?	Feasibility report, geotechnical investigation, ESIA submission, grid study
End of permitting	Does the project hold the rights it needs?	Licences, water permit, land agreements, environmental approval
End of PPA stage	Is there a financeable offtake?	Signed or initialled PPA and implementation agreement, regulator approval
Financial close	Are all conditions precedent satisfied?	The 23 gates of Chapter 17

The most important gate is the one at the end of pre-feasibility, because the next stage costs several times more than everything spent so far. Kasiri's feasibility stage costs USD 4.5 million against USD 0.7 million spent before it. A developer that enters feasibility without a credible PPA route and a defensible energy estimate is spending most of its development budget on hope.

5.5 The probability-weighted timeline

Each stage that might fail also might be late. Lenders and buyers plan on a commercial operation date; the developer should plan on a distribution of dates. A simple discipline is to report, alongside the target date, the date by which the project has at least an even chance of closing if it survives, and to update both at every gate. The model does not simulate delay in development; it treats development as a fixed sequence ending at financial close in 2027. The value of time is captured in Chapter 14 through the developer's discount rate.

5.6 What the Kasiri case shows

Tamarind has completed feasibility and is midway through permitting and PPA negotiations. Its remaining development budget is mostly the financing stage, and its remaining risks are concentrated in land rights, the regulator's approval of the tariff and the lenders' view of the flow record. That is a typical profile for a project at this point, and it explains the "ask" in Chapter 18: a co-developer to fund the rest of development and a DFI-led lender group.

Points for the investment and credit committees

1. Ask for the stage plan with, for each stage, the questions, the evidence, the budget, the duration and the decision-maker, agreed before the stage starts.
2. Treat the end of pre-feasibility as the main go or stop decision, because the next stage costs several times more than everything before it.
3. Read stage probabilities as model inputs that make the economics explicit, not as statistics.
4. Ask how many projects the developer has started for each one it closed, and compare with the probabilities in its plan.
5. Report a probability-weighted timeline, not only a target date.

Working with the model

01A_DEVELOPMENT holds the six stages with their durations, budgets and probabilities, the spend profile by year, the probability that the project is still alive when each dollar is spent, and the value of the development position at the start of each stage. Change a probability and read the risk-weighted developer NPV on the same sheet and on *32_DASHBOARD*.

Chapter 6. The development budget and how it is funded

This chapter supports the decision on how much to spend on development, in what order, and with whose money. The development budget is small relative to the construction budget, but it is spent when the project is worth least and the risk of loss is highest. How it is funded decides who owns the project at close.

Place in the analytical chain: the development plan, continued; the money that buys the option described in Chapter 1.

6.1 What development costs

Public data on development budgets for African hydro are thin. The evidence that exists is scattered across scales. The IFC's guide puts site identification at about USD 10,000 to 20,000 for a small hydro site and USD 150,000 to 200,000 for a larger one [DE:S1]. Feasibility and ESIA work for two small sites of 2 to 3 MW cost about USD 227,000 [DE:S14]; for a 1 MW site in Liberia, including design, about EUR 600,000 [DE:S15]; for a project of about 53 MW in Tanzania with its line, about USD 4.8 million [DE:S16]. Castalia, reported in Engineering News, put total development cost for a utility-scale renewable project in Africa at USD 1 million to 2.5 million [DE:S18], a range that reflects solar and wind more than hydro. An IFC InfraVentures presentation implies development budgets of roughly 7 to 10 percent of capital cost for large projects, a figure derived from its typical ticket and share [DE:S2].

Kasiri's development budget is USD 9.0 million in 2026 dollars, about 5.7% of plant cost. The largest item is the feasibility stage, which carries the geotechnical investigation, the ESIA and the grid study.

6.2 At-risk and recoverable spend

Development spending is recovered only if the project closes. At close, lenders normally allow the project company to reimburse the developer's documented development costs out of the financing, within a cap agreed in the term sheet, and the developer receives a development premium on top. If the project fails, the money is lost.

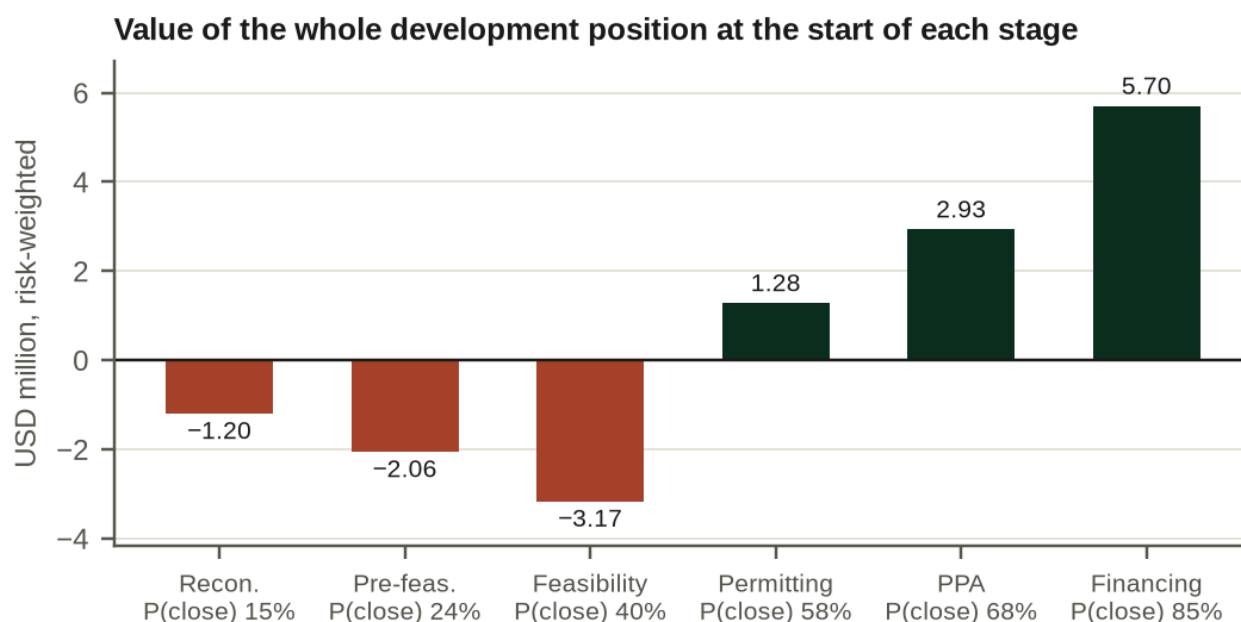
That makes the timing of spend as important as its size. Spending that can be deferred to a stage with a higher probability of success is worth more than spending that cannot. Kasiri's model shows the effect in one row: the probability that the project is still alive when each dollar is spent. In the year the feasibility study starts, that probability is about 40%; in the year before close, about 17%.

6.3 Expected spend and the value of the position

Weighted by those probabilities and discounted at the developer's 25 percent rate, Kasiri's development spend has a present value at the start of development of about USD 1.38 million, against USD 3.95 million if every stage were certain to succeed.

The same logic values the development position at the start of each stage. That value is what a co-developer should be willing to pay to buy in, before negotiation.

Stage about to start	P(close) from here	Risk-weighted value of the position (USDm, 100%)
1. Reconnaissance	15%	-1.20
2. Pre-feasibility	24%	-2.06
3. Feasibility	40%	-3.17
4. Permitting	58%	1.28
5. PPA	68%	2.93
6. Financing	85%	5.70



The table answers a practical question. A position that is worth little or less than nothing at reconnaissance becomes valuable as the probability of close rises and the remaining spend falls. A co-developer who enters before feasibility takes more risk and should get a larger share for its money than one who enters at the financing stage. In the base case the value is negative until the feasibility study has been paid for, and most negative just before it, because the largest single spend is still ahead and the developer's return at close is below its 25 percent cost of capital. It turns positive at permitting, once that spend is sunk and its reimbursement at close is in view. Chapter 14 shows how the premium and the tariff change that.

6.4 Who funds development

Five sources fund development in Africa.

The developer's own equity funds the early stages almost everywhere. It is the most expensive money, and the developer needs a portfolio of projects to absorb the failures.

Co-developers and development finance institutions buy in, usually after pre-feasibility. IFC InfraVentures funds an average of about USD 4 million per project, for a 20 to 30 percent share of development costs, according to a 2015 presentation [DE:S2]. Africa50's project development arm invests about USD 2 million to 10 million in projects above USD 100 million [DE:S21].

Project preparation grants fund studies. The African Development Bank's Sustainable Energy Fund for Africa granted about USD 1 million to prepare a community-owned hydro project in Kenya [DE:S17]. Grants lower the developer's cost of capital but often come with conditions on procurement and ownership.

Utilities and governments fund development of schemes they intend to own, or to tender. A tendered site, prepared by the state, shifts the development risk away from private bidders and changes the economics of Chapter 14 entirely.

Contractors sometimes fund development in exchange for the construction contract. This is fast, but it removes the competitive pressure on construction cost that lenders rely on, and it should be read as a price, not a gift.

6.5 What the Kasiri case shows

Tamarind has funded about USD 5.2 million of development so far from its own balance sheet. The remaining USD 3.8 million covers permitting, the PPA and the financing stage, and is what it wants a co-developer to provide. The value table shows that a buyer entering at the start of permitting would be paying for a position that is worth about USD 1.28 million

on the developer's assumptions. That number, not the money already spent, is the starting point for the negotiation in Chapter 18.

Points for the investment and credit committees

1. Ask for the development budget by stage and workstream, distinguishing at-risk spend from spend that will be reimbursed at close, and the cap on reimbursement in the term sheet.
2. Value the development position at the stage of entry, using explicit probabilities and a stated discount rate, before negotiating a share.
3. Defer spending to later, more certain stages where the critical path allows it.
4. Read contractor-funded development as a price paid in construction cost.
5. Record grants and their conditions, because they can restrict procurement and ownership later.

Working with the model

01A_DEVELOPMENT section A holds the budget by stage; section C spreads it over the development years; section D values the developer's position; section D2 values it at the start of each stage. The development costs reimbursed at close flow into plant cost on *05_PLANT_CAPEX*, and the development premium into the funding base on *17_PROJECT_FINANCE*.

Chapter 7. Consents, land and E&S: the permitting sequence

This chapter supports the decision on which consents to pursue first and which sit on the critical path to financial close. Permits are not a list to be completed; they are a network in which some approvals depend on others, and in which the slowest item sets the date of close.

Place in the analytical chain: consents, the legal rights that turn a feasible project into a developable one.

7.1 The map of consents

A hydro project in Sub-Saharan Africa typically needs most of the following. The names and the authorities differ between countries; the dependencies do not.

Consent	Typical authority	Depends on	Why lenders care
Water use permit or water right	Water resources authority	Hydrology study, environmental flow	Without it, the project has no legal right to the river
Environmental and social approval	Environment agency	ESIA, public consultation	Condition of lending under international standards
Generation licence	Electricity regulator	Feasibility, water permit, environmental approval	Required to sell power
Land rights and resettlement	Land authority, local government, communities	Resettlement action plan, compensation	Access for construction; reputational risk
Concession or implementation agreement	Government (energy and finance ministries)	Licence, PPA	Defines state obligations, including termination
Grid connection agreement	Transmission company or utility	Grid study	Evacuation and deemed energy
Dam safety approval	Dam safety authority, where one exists	Design review	Lenders apply international dam safety practice [HY-15]
Tax, customs and exchange control approvals	Revenue authority, central bank	Investment registration	Cash flow and repatriation

The water permit and the environmental approval usually come first, because the licence and the concession depend on them. Land is often the slowest item and the one most likely to delay construction after close.

7.2 Durations

Published durations are scarce. Uganda's electricity regulator has up to 180 days to process a generation licence application, according to a secondary source [DE:S5]. Environmental approvals depend on the scale of the project and the quality of the ESIA. Land acquisition can take years where customary rights, graves or informal settlements are involved.

The practical rule is to identify the consent with the longest expected duration and the highest chance of failure, start it first, and avoid spending on later stages until it is secure.

7.3 Lender standards

International lenders apply environmental and social standards that are stricter than many national laws. The Equator Principles apply to project financings with a total capital cost of USD 10 million or more, and require projects in Sub-Saharan Africa to comply with the IFC Performance Standards [DE:S49; DE:S50]. The Hydropower Sustainability Standard provides a sector-specific framework that some lenders and buyers ask for [HY-03]. In practice the lenders' environmental and social adviser will review the ESIA against those standards and agree an environmental and social action plan, which becomes a condition of disbursement.

A national approval is therefore necessary but not sufficient. Kasiri's ESIA has been approved nationally but not yet reviewed against lender standards, so Gate 5 of the readiness sheet reads NOT MET.

7.4 Resettlement

Resettlement is the consent most likely to delay construction and the one with the largest reputational risk. Lenders expect a resettlement action plan that is approved, budgeted and implemented before the land is needed. At Rusumo Falls, land acquisition and livelihood restoration costs roughly doubled during implementation, from about USD 18 million to about USD 38 million [A2:S16].

Kasiri's resettlement action plan has been approved but not yet implemented, and land rights for the tunnel portal and the powerhouse are not yet secured. That is the item on its critical path.

7.5 Rivers that cross borders

A river shared with another country brings a separate set of obligations: notification of riparian states, and in some cases their agreement. Lenders bound by safeguard policies treat unresolved transboundary issues as a reason not to lend. The Grand Ethiopian Renaissance Dam was completed without a binding operating agreement with Egypt and Sudan [A1:S1], which illustrates how long such questions can outlast construction. Kasiri's river lies entirely within Navaria.

7.6 What the Kasiri case shows

Kasiri's critical path runs through land and the lender-standard review of its ESIA, not through the licence or the PPA. The developer should secure land rights for the main works before it spends the bulk of the financing-stage budget on lenders' advisers.

Points for the investment and credit committees

1. Ask for the consent map with dependencies, owners and expected dates, and identify the critical path.
2. Treat national environmental approval and lender-standard compliance as two separate gates.
3. Require a resettlement action plan that is approved, budgeted and scheduled against construction access dates.
4. Check that permit tenors cover the debt tenor plus a tail.
5. Ask whether the river crosses a border, and what notification or agreement has been obtained.

Working with the model

13_REGULATION holds the regulatory readiness matrix and the E&S readiness inputs that feed Gate 9 of *30_BANKABILITY* and Gates 5 to 8 of *30A_CLOSE_READINESS*. *29_RISK_ALLOCATION* records who carries E&S risk.

Chapter 8. The contract set: concession, PPA, grid connection and direct agreements

This chapter supports the decision on whether the chain of contracts is bankable, consistent and complete. Lenders lend against contracts, not against concrete. A gap or an inconsistency between two contracts is a risk that nobody has agreed to carry, and it will be found in due diligence or, worse, after close.

Place in the analytical chain: the contract set, which turns the consents and the offtake route into enforceable obligations.

8.1 The contract map

A hydro IPP sits at the centre of about ten contracts.

Contract	Parties	What it must do
Concession or implementation agreement	Government and project company	Grant the right to build and operate; set state obligations, change in law, termination and compensation
Power purchase agreement	Utility and project company	Set tariff, indexation, currency, volume terms, payment security, force majeure, termination
Grid connection agreement	Transmission company and project company	Set connection works, responsibility, dates and consequences of delay
Water use permit (or agreement)	Water authority and project company	Set the right to abstract, environmental flow and conditions
EPC contract(s)	Contractors and project company	Set price, completion date, performance, liquidated damages, security
O&M contract	Operator and project company	Set availability, costs, incentives
Shareholders' agreement	Sponsors	Set funding obligations, governance, transfers
Financing agreements	Lenders and project company	Set debt terms, covenants, security, conditions precedent
Direct agreements	Lenders and each key counterparty	Give lenders step-in rights if the project company defaults
Guarantees and insurance	Guarantors, insurers, project company, lenders	Cover risks no counterparty can carry

8.2 Consistency

The most common defects are inconsistencies between contracts. Four checks catch most of them.

Durations. The PPA, the concession, the water permit and the land rights should all run at least as long as the debt, with a tail. Kasiri's assumed 25-year operating term is longer than its 20-year PPA in the case description, an inconsistency the book flags deliberately: the model assumes a five-year extension at the same tariff, and a lender would want that extension written into the contracts or excluded from the base case.

Force majeure. An event that excuses the buyer from paying under the PPA must also excuse the project from its obligations under the EPC and financing contracts, or the project carries the gap.

Termination amounts. If the PPA or the concession can be terminated for government default or political force majeure, the compensation should at least repay the senior debt outstanding. The model computes the termination exposure each year as debt outstanding plus the larger of unrecovered private equity with a 15 percent premium and the present value of the distributions equity would still have received. At Kasiri's commercial operation date that exposure is about USD 216 million.

Grid interface. The PPA's deemed-energy clause and the grid connection agreement should agree on who pays when the line is late. Where the project builds its own interconnector, as Kasiri does, the risk sits with the project and its construction contract.

8.3 Guarantees and their counter-indemnities

Where the buyer's credit is weak, lenders often ask for a guarantee. Multilateral guarantees to commercial lenders are backed by indemnities from the host government. Uganda signed an indemnity agreement with IDA in 2007 to support the Bujagali partial risk guarantee of up to USD 115 million [A2:S53]. Cameroon backs the IBRD guarantees for Nachtigal through a similar indemnity [A2:S3]. A guarantee therefore changes the lender's counterparty without removing the state's obligation, which is why Chapter 15 counts these instruments as contingent liabilities.

8.4 What the Kasiri case shows

Kasiri's contract set is incomplete: the PPA and implementation agreement are agreed in principle but not signed, land agreements are outstanding, and the construction contracts are at term-sheet stage. The two consistency issues to resolve before close are the gap between the PPA and the operating term, and a termination amount in the implementation agreement that covers the debt.

Points for the investment and credit committees

1. Ask for the contract map and a consistency checklist covering durations, force majeure, termination and the grid interface.
2. Confirm that termination compensation for government default at least repays senior debt.
3. Require direct agreements with every key counterparty.
4. Record every guarantee with its counter-indemnity, and treat it as a state obligation.
5. Exclude from the base case any revenue that depends on a contract term not yet written.

Working with the model

14_PPA holds the PPA terms and the termination premium. *23_GUARANTEES* computes debt-guarantee, PPA-guarantee, termination and currency exposures year by year; *24_CONTINGENT_LIABILITIES* combines them without double counting. *17A_STRUCTURES* records which structures carry a sovereign PPA guarantee and a termination obligation.

Chapter 9. EPC and construction contracting

This chapter supports the decision on how to contract the construction, and therefore who takes ground and delay risk, at what price. In hydro the contract structure is a financing decision, because the largest cost risk is underground and lenders will size their comfort to whoever carries it.

Place in the analytical chain: construction, the second of the three investments.

9.1 Three structures

There are three main ways to contract a hydro plant.

A single turnkey engineering, procurement and construction (EPC) contract makes one contractor responsible for the whole plant, at a fixed price and a fixed completion date, backed by liquidated damages and bonds. It buys price and date certainty, and the contractor charges for it.

Split packages divide the work into a civil contract and an electro-mechanical contract, sometimes with a separate hydro-mechanical package. Each contractor carries its own scope, and the owner carries the interface between them.

Multi-contracting divides the work further, with the owner, through its engineer, integrating many packages. It is the cheapest on paper and leaves the most risk with the owner.

The model compares the three with four parameters, all assumptions for comparison, to be replaced by bids.

Parameter	Turnkey EPC	Split packages	Multi-contract
Contractor risk premium on civil, hydro-mechanical and electro-mechanical prices	12%	6%	0%
Share of a cost overrun borne by the owner	30%	55%	90%
Share of delay cost borne by the owner after liquidated damages	35%	60%	90%
Interface risk	Low	Medium	High

9.2 Ground risk

The label "fixed price" is misleading in hydro unless the contract says how unforeseen ground conditions are treated. Many turnkey contracts exclude them or cap the contractor's exposure, so the owner carries the largest risk whatever the contract is called. FIDIC's Emerald Book, published in 2019 for underground works, makes the geotechnical baseline report the sole contractual description of the expected ground: the contractor carries ground conditions within the baseline, the employer carries conditions beyond it, and time for completion is adjusted accordingly [DE:S47].

The consequence for the developer is that the quality of the geotechnical investigation sets the price of the contract. A thin investigation produces a baseline that the contractor will price defensively and a long list of exclusions. The consequence for lenders is that a turnkey label should be read through to the treatment of ground risk before any credit is given for it.

9.3 Payment terms, security and damages

The IFC's guide reports typical down payments of 10 to 30 percent of the contract price and retentions of 10 to 20 percent; equipment is often paid 30 percent in advance, 50 percent on delivery and 20 percent on acceptance [DE:S1]. Down payments raise the project's early funding need and interest during construction. Retentions and performance bonds give the owner a hold over the contractor until completion.

Liquidated damages for delay should cover at least the project's lost revenue and interest during the delay; damages for performance shortfall should cover the value of lost energy over the life of the plant. Caps on damages, typically a share of the contract price, decide how much of the risk really passes to the contractor.

9.4 Contractor-financed packages

Several large African plants were built under an EPC contract bundled with export-credit finance arranged by the contractor. Isimba and Karuma in Uganda were 85 percent financed by China Exim [A2:S34; A2:S40]. Such packages close quickly, but they weaken competitive pressure on price and can weaken the owner's control of quality. At Isimba, several hundred defects were reported after commissioning, and at Karuma the contractor sought about USD 40 million to fix defects [A2:S34; A2:S42]. An independent owner's engineer and retentions sized to the risk are the minimum protection.

9.5 What the model shows

The model runs the three structures with debt sized in each case, first in the base case and then with an overrun at the reference-class mean of 96 percent.

Run	Turnkey EPC	Split packages	Multi-contract
Plant cost, base (USDm, real)	164	157	150
Private equity IRR, base	12.6%	13.8%	14.9%
Private equity IRR, 96% overrun	7.4%	5.6%	3.8%
Financing gap, 96% overrun (USDm)	43	65	96

The pattern is the point. Multi-contracting gives the best return when nothing goes wrong and the worst when something does; turnkey EPC gives the reverse. The choice is a trade between expected cost and the shape of the downside, and lenders, who do not share the upside, will prefer the structure with the narrower downside unless the owner has the balance sheet and the engineering capacity to absorb overruns.

9.6 The owner's team and the lenders' adviser

Whatever the structure, the owner needs an engineer who represents it on site, checks designs, certifies payments and manages claims. The lenders appoint their own technical adviser, who reviews the design, the contracts, the schedule and the budget before close and monitors construction afterwards. The two roles should not be combined.

Points for the investment and credit committees

1. Read every "fixed-price" contract through to its treatment of ground conditions, and ask for the geotechnical baseline.
2. Compare structures on the downside, not only on price: the owner's share of an overrun and of delay.
3. Check that delay damages cover lost revenue and interest, and that caps leave the contractor with real exposure.
4. Treat contractor-arranged finance as a price paid in reduced competition, and require an independent owner's engineer.
5. Size retentions and bonds to the risk, not to convention.

Working with the model

05A_CONTRACTING selects the structure and holds its parameters. The contractor's premium flows into *05_PLANT_CAPEX*; the owner's share of an overrun and of delay costs flows into the capital cost on *06_CONSTRUCTION* when the overrun or delay stress is switched on. The six contracting runs are in the full-engine snapshot.

Chapter 10. Construction budget, contingency and funding the build

This chapter supports two decisions: how large the contingency should be, and how construction will be funded if costs rise. A contingency sized by convention and a financing plan that assumes it will be enough are the two most common reasons why a hydro project that closed comfortably is in default before it is commissioned.

Place in the analytical chain: construction, continued; the budget and the money behind it.

10.1 Total uses

Lenders and sponsors fund total uses, not plant cost. For Kasiri, in nominal terms:

Use	USDm
Plant capital cost, escalated	165
Project-built transmission line	18
Development premium paid to the developer at close	4.7
Funding base	187
Interest during construction	17
Upfront financing fees	2.9
Initial debt service reserve	7.3
Total uses	215

The plant cost already includes the development costs reimbursed at close and a 10 percent physical contingency.

10.2 Contingency

The IFC's guide reports a median contingency of 9.8 percent of total plant cost across its sample, with a range of 5.4 to 12.6 percent, and an industry practice of about 15 percent for civil works and 7.5 to 10 percent for electro-mechanical and grid works [DE:S1]. A contingency set by item, with the larger share on civil works and tunnels, is more defensible than a single percentage.

The reference class gives a sobering comparison. Across 245 large dams, the median real cost overrun was 27 percent and the mean 96 percent [HY-08]. That study covers large dams, so it overstates the risk for a medium run-of-river plant with a short tunnel; it is still the best public reference class available, and it shows that a 10 percent contingency is a central estimate, not a margin of safety.

10.3 What an overrun does

With the financing package locked at its base-case terms, the model tests two overruns, applied to the owner's share under the split-package structure.

Run	Financing gap (USDm)	Private equity IRR	Minimum DSCR
Base, financing locked	4	13.8%	1.53x
Overrun at the reference-class median (27%)	22	10.2%	1.52x
Overrun at the reference-class mean (96%)	68	5.7%	1.49x
Two-year delay	24	10.1%	1.55x

With the debt locked, an overrun falls on equity. The DSCR barely moves, because debt service is fixed; the financing gap grows, because someone has to pay for the extra cost. That gap is the number that matters. Unless the sponsors have committed standby equity, or the lenders have provided a standby facility, an overrun of that size stops construction.

10.4 Funding overruns

There are four ways to fund an overrun, and a bankable project has at least two of them in place at close.

Contingency inside the budget absorbs the expected overrun. Standby equity, committed by the sponsors at close and drawn only if needed, absorbs the next layer. A standby debt facility, provided by the lenders at a higher margin, absorbs a further layer within the lenders' coverage tests. Completion support, a guarantee by the sponsors to fund whatever is needed to reach commercial operation, transfers the residual risk to the sponsors' balance sheets until completion.

A developer without the balance sheet to give completion support should expect lenders to ask for more contingency, more standby equity or a turnkey contract.

10.5 Interest during construction and the draw schedule

Interest during construction depends on when money is drawn. Equity first, debt later, reduces interest during construction and is what lenders usually require; drawing debt and equity pro rata raises it. The model draws both pro rata with spending, on an annual basis, which slightly flattens the equity return; a monthly, equity-first draw schedule is on the list of extensions in Annex K.

Interest during construction is funded by debt within the gearing limit. To avoid circularity, the model uses a closed form: if debt is drawn in proportion to spending, interest per dollar of debt is a constant k set by the drawdown profile and the rate, and the debt that respects a gearing limit g with fees f is

$$D = g \times K / (1 - g \times (k + f))$$

for a single tranche, with an extension for a concessional tranche. This is a standard technique, not an innovation; its advantage is that the model needs no macro and no iteration.

Points for the investment and credit committees

1. Ask for the contingency by item, with the basis for each, and compare it with the reference class.
2. Test overruns with the financing package locked, and read the financing gap, not only the DSCR.
3. Require at least two layers of overrun funding at close: contingency and standby equity or debt.
4. Ask for the draw schedule and the order of drawdown, and its effect on interest during construction.
5. Read the reference class as a warning for medium plants, not as a calibrated forecast.

Working with the model

17_PROJECT_FINANCE shows sources and uses, the gearing and the financing gap. *06_CONSTRUCTION* holds the spending profile and escalation. *18_DEBT* shows the drawdown and the closed-form interest factors. Switch on the overrun and delay stresses on *01_CONTROL_PANEL*, with *debt_mode* set to 2 to lock the financing package.

Chapter 11. Operations: availability, O&M, major maintenance and insurance

This chapter supports the decision on how the plant will be operated and maintained, and how to reserve for the large, irregular costs that come years after commissioning. Operations receive little attention in financing negotiations because hydro is cheap to run. That is exactly why operational failures are underestimated.

Place in the analytical chain: operations, the start of the third investment.

11.1 Who operates the plant

Three models are common. The owner operates the plant with its own staff, which keeps control and knowledge but requires a team the developer may not have. An operations and maintenance contractor runs the plant under a contract with availability targets and incentives. The equipment manufacturer provides a long-term service agreement for the turbines and generators, usually alongside one of the first two. Lenders care less about the model than about the incentives: who pays when the plant is unavailable.

11.2 Costs

IRENA reports fixed operation and maintenance costs for hydro of 1 to 3 percent of installed cost a year, with an average a little under 2 percent and lower ratios for large plants [HY-01]. The IFC's guide gives 1 to 4 percent for small hydro [DE:S1]. Kasiri's operating costs in its second year of operation, including fixed and variable O&M, insurance, a major maintenance reserve, administration and community programmes, are about USD 6.1 million. The water-use royalty, a payment to the state, adds about USD 0.8 million.

11.3 Availability

The model assumes 95 percent availability after the first year and a commissioning ramp in the first. Under an energy-only tariff, every point of unavailability is lost revenue; under a two-part tariff, it reduces the capacity payment. Availability guarantees from the O&M contractor and the equipment supplier should be read against that exposure.

11.4 Major maintenance

Turbines need major overhauls, typically at intervals of a decade or more, and civil works need periodic repair. The costs are large and irregular. Lenders usually require a major maintenance reserve account funded from cash flow before distributions. The model funds a reserve contribution as an operating cost each year, a simplification that smooths the cost; a full model would schedule overhauls and draw the reserve.

Post-commissioning civil problems are not rare. At Reventazón in Costa Rica, spillway foundation repairs estimated at USD 15 million were needed after commercial operation [INT:S22]. World Bank guidance on operation and maintenance strategies for hydro sets out how to plan for such costs [HY-07].

11.5 Insurance

Construction all-risks and delay-in-start-up insurance protect the project during construction; operational all-risks and business interruption cover protect it afterwards. Hydro plants are exposed to floods, landslides and sediment, and insurers price those risks by site. The model carries insurance as a percentage of capital cost; the insurance adviser's report replaces it before close.

Points for the investment and credit committees

1. Ask who pays for unavailability under each operating contract, and compare with the tariff's exposure.

2. Require a major maintenance plan and a reserve funded before distributions.
3. Check that the insurance programme covers flood, landslide and sediment risks at the site, and delay in start-up.
4. Budget for civil repairs in the early years of operation.
5. Treat low operating cost as a reason for discipline, not for neglect.

Working with the model

07_OPEX holds operating cost inputs and the annual cost lines, including the water royalty. *03_HYDROLOGY* holds availability and the commissioning ramp. *18_DEBT* and *20_CASH_FLOW* hold the debt service reserve; the major maintenance reserve is carried as an operating cost on *07_OPEX*.

Chapter 12. Sizing hydro debt: the lender case

This chapter supports the decision on how much senior debt the project can carry, at what tenor and on which generation case. The amount of debt sets the amount of equity, and the amount of equity, at the developer's cost of capital, sets the tariff the project needs. Debt sizing is therefore not a technical step after the commercial negotiation; it is part of it.

Place in the analytical chain: debt, the first link of the financing layer.

12.1 Tenor

For an asset that can run for fifty years or more, the length of the debt matters more for the tariff than almost any other financing term. Commercial tenors for African hydro are rarely longer than fifteen to eighteen years [PF-13]. Kikagati in Uganda closed with a 16-year loan from FMO and the Emerging Africa Infrastructure Fund [DE:S53]. Bugoye was refinanced in 2017 with 12-year debt [DE:S30]. Nachtigal raised an 18-year euro tranche and a 21-year local-currency tranche guaranteed by IBRD [PF-11].

Kasiri's senior debt has a 16-year repayment period after commercial operation at 8 percent, fixed or swapped for three quarters of the amount, with a target DSCR of 1.35 for sizing.

12.2 Gearing and cover

Public evidence on gearing for small and medium African hydro sits between about 62 and 76 percent of total cost: about 70 percent at Nyamagasani 1, 75 percent at Nyamagasani 2 and about 62 percent at Kikagati (derived) [DE:S39b; DE:S39c; DE:S53]. The IFC's guide indicates DSCR targets of 1.2 to 1.5, with cover above 1.0 in the worst dry year [DE:S1]. The most specific public covenant evidence for a large African hydro IPP is Nachtigal's lenders' base case, with a minimum DSCR of 1.45 and an average of 1.53 [PF-11].

Debt is sized by whichever constraint binds first: the gearing limit or the cover requirement. For a project with an energy-only tariff and a variable river, cover usually binds. At Kasiri the two constraints nearly coincide: the lender case on the ten-year P90 would support about USD 148 million at a DSCR of 1.35, and the 70 percent gearing limit, applied to capital cost plus debt-funded interest and fees, allows about USD 145 million, about 68% of total uses. On the one-year P90, cover binds by a wide margin, as section 12.4 shows.

12.3 Sculpting

The model sizes debt by sculpting debt service to the target DSCR on the lender case:

$$DS_k = CFADS_{\text{lender}, k} / DSCR_{\text{target}}$$

$$\text{Debt capacity} = \sum_k DS_k / (1 + r)^k$$

where CFADS is cash flow available for debt service in loan year k and r the interest rate. Sculpting matches debt service to the shape of the cash flow, which suits a hydro plant whose output is roughly flat in real terms.

12.4 Which generation case

The lender case uses a downside generation case, and the choice changes the debt. Chapter 2 distinguished the one-year P90 from the P90 of the ten-year average. Sizing on the one-year P90 for every year is the most conservative choice; sizing on the ten-year P90 is more generous and needs a separate test for a multi-year drought. The model's default sizes on the ten-year P90 and tests a three-year drought explicitly.

Lender case	Debt capacity (USDm)	Gearing on total uses
P90 ten-year average (default)	145	68%
P90 one-year, every year	127	60%

The difference is equity, which is the most expensive capital in the structure. The debate between sponsor and lender over the lender case is a debate about who funds that difference.

12.5 Reserves

A debt service reserve account holds a number of months of debt service, funded at close and drawn when cash falls short. The model holds six months, draws the reserve first in a shortfall, releases only the excess over the target and tops it up only from available cash. In Kasiri's three-year drought, the reserve is drawn and then rebuilt; the drought still pushes the minimum DSCR to 0.69x, below 1.0, which means that the reserve alone is not enough. A larger reserve, a cash sweep in wet years or a lender case sized on the one-year P90 would each address it at a cost in equity.

12.6 Results

Table 12.1. Kasiri River Hydro: base-case debt and coverage (debt sized in the model)

Metric	Value
Total uses	USD 215 million
Senior debt	USD 145 million
Gearing on total uses	68%
Minimum and average DSCR, P50 case	1.53x and 1.60x
LLCR at commercial operation	1.59x
Project IRR, post-tax, nominal	11.2%
Private equity IRR, nominal	13.8% against a 15.0% target
Financing gap	USD 4.1 million

The minimum DSCR in the P50 case is above the 1.35 sizing target because debt is sized on a downside case. The equity return is below target, and private equity required is slightly above what the sponsors have committed: the gap is small, but it is a gap.

12.7 Testing a package already sized

Once a package is sized, lenders ask how it performs when things go wrong, not how much they would lend in a bad world. The model's locked mode holds every tranche amount, the grant, the government's equity and the commercial principal schedule at their base-case values; interest floats only on the unhedged share, and equity absorbs the difference. All stress results in this book use that mode.

Points for the investment and credit committees

1. Ask which constraint binds, gearing or cover, and on which generation case.
2. Agree explicitly whether the lender case uses the one-year or the ten-year P90, and how a multi-year drought is tested.
3. Check that the debt service reserve, alone or with a cash sweep, covers a realistic drought.
4. Test stresses with the financing package locked.
5. Compare the tenor with the asset life, and ask what a longer tenor would do to the tariff.

Working with the model

18_DEBT holds the sculpting, the tranches, the interest factors and the debt service reserve. *17_PROJECT_FINANCE* holds sources and uses, debt capacity, gearing, the financing gap and the main ratios. *20_CASH_FLOW* holds the waterfall, the DSCR row and the reserve movements. The lender case is selected on *01_CONTROL_PANEL*.

Chapter 13. The lender group: DFIs, blended finance and credit enhancement

This chapter supports the decision on which lenders to approach, which guarantees and insurances to seek, and at what cost to whom. The lender group decides tenor, price and conditions; the credit enhancement decides who carries the risks that the lenders will not.

Place in the analytical chain: the lender group, between the debt sizing and the developer's returns.

13.1 Who lends to African hydro

Development finance institutions lend most of the senior debt for private African hydro, often in clubs. Nachtigal's debt came from eleven development finance institutions and local banks, coordinated by IFC [A2:S6]. Small plants in Uganda were financed by FMO, the Emerging Africa Infrastructure Fund and Proparco, among others [DE:S39; DE:S53]. Commercial banks participate mainly through A/B loan structures, in which a development finance institution is the lender of record, and through local-currency tranches.

Export credit agencies finance equipment and sometimes whole plants, usually tied to a contractor from their country. Their terms can be long and cheap, and they come with the contractor.

13.2 Local currency

A local-currency tranche reduces the currency mismatch for a project paid in local currency. Nachtigal's 21-year local-currency tranche, guaranteed by IBRD, is the leading African example [PF-11]. For a project paid in dollars, as Kasiri is, local-currency debt adds currency risk to the project rather than removing it; the mismatch sits with the utility instead, and Chapter 15 measures it.

13.3 Concessional and blended finance

Concessional debt, grants and results-based premiums lower the cost of capital and therefore the tariff. The model compares five structures for the same plant. The full engine solves, for each structure with private equity, the tariff that gives the sponsors their target return.

Structure	Private equity IRR at the base tariff	Tariff for target return (USD/MWh, 2026)
IPP with DFI debt (base)	13.8%	116
PPP with concessional debt and viability gap funding	22.3%	92
Hybrid, public civil works	45.1%	69
Blended finance	26.2%	83

The table carries the most important rule of blended finance. At the same tariff, a grant raises the sponsors' return: in the hybrid structure, where the state funds 35 percent of the cost, the private equity return at the base tariff is 45.1%. A grant or concessional tranche that is not matched by a lower tariff or a cap on returns transfers public money to private shareholders. The right use of concessional money is to lower the tariff to the level in the last column.

Uganda's GET FiT programme is an example of results-based support designed with that in mind: a premium of about USD 0.014 per kWh for small hydro, calculated on 20 years of output but paid within five years, half at commercial operation [DE:S9], awarded alongside a regulated tariff.

13.4 Credit enhancement

Four instruments cover risks that lenders will not take.

Political risk insurance from MIGA covers expropriation, war, transfer restriction and breach of contract, and MIGA's non-honouring cover protects against a sovereign's failure to pay a financial obligation [PF-03].

Partial risk and loan guarantees from the World Bank and the African Development Bank cover lenders against a government's failure to meet specific obligations, backed by an indemnity from the government [PF-01; PF-02; PF-06].

Liquidity facilities bridge temporary payment delays by the buyer. ATIDI's Regional Liquidity Support Facility can cover up to twelve months of revenue for projects of up to about 100 MW, according to its published description [PF-04].

Letters of credit and escrow accounts provided by the buyer cover a number of months of billing.

Each of these moves risk towards the state or towards an institution the state indemnifies. A liquidity facility is the partial exception, because it bridges delays rather than absorbing losses.

13.5 Refinancing

Once a plant is built and has a record of operation, its debt can be refinanced on better terms. Bugoye in Uganda was refinanced in 2017 by the Emerging Africa Infrastructure Fund and FMO, after its original sponsors had sold [DE:S30]. Refinancing allows the developer or its buyer to release equity and recycle it into new projects, and it is part of the value step-up between close and operation discussed in Chapter 14. The model does not refinance; the step-up is valued directly.

Points for the investment and credit committees

1. Ask which lenders are in the group, on what terms, and which are lenders of record for commercial banks.
2. Match the currency of the debt to the currency of the revenue, and measure where the mismatch lands.
3. Require every grant or concessional tranche to be matched by a lower tariff or a cap on returns.
4. Record every guarantee and its counter-indemnity.
5. Plan for refinancing after operation, but do not rely on it in the base case.

Working with the model

17A_STRUCTURES holds the five structures, the closed-form screen and the full-engine snapshot with the tariff each structure needs. *23_GUARANTEES* and *24_CONTINGENT_LIABILITIES* record the exposures created by each instrument. *11_OFFTAKER* holds the payment security.

Chapter 14. Developer returns, valuation and sell-down

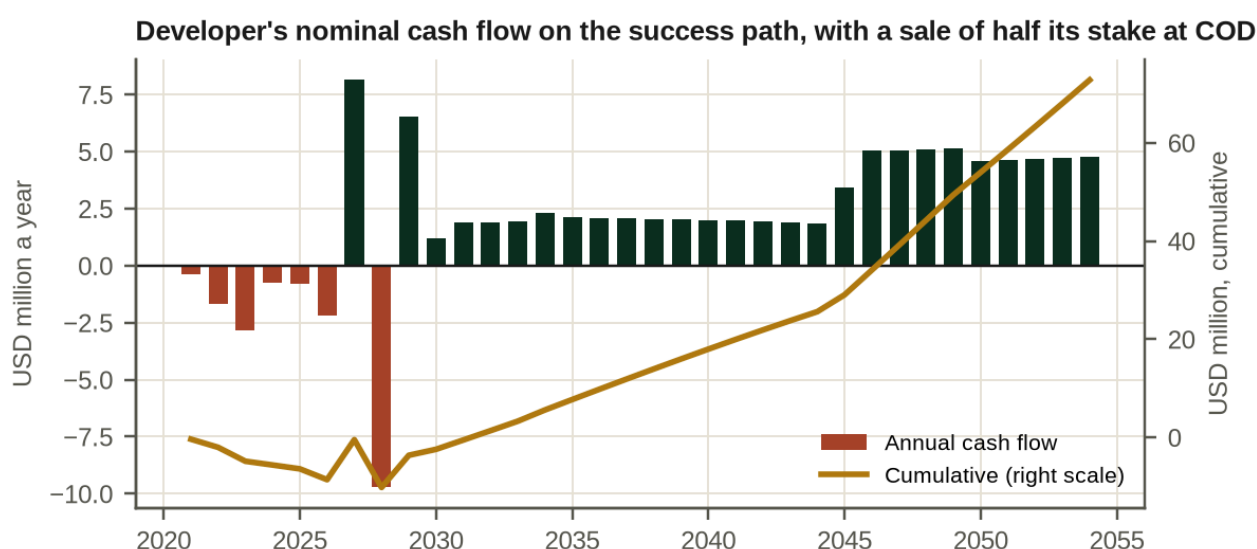
This chapter supports the decision on whether the project is investable for the developer, and at what price and stage the developer should bring in partners or sell. It is the chapter most often missing from hydro analysis, because the equity return at close is easy to compute and looks like the developer's return. It is not.

Place in the analytical chain: developer returns, which close the financing layer from the sponsor's side.

14.1 Three measures of the developer's return

The developer's economics need three measures, and each answers a different question.

The success-path return follows the developer's cash from the first study to the last distribution, assuming the project closes. It includes development spending, the reimbursement of development costs at close, the development premium, the developer's share of equity contributions and distributions, and any sale of part of its stake. For Kasiri, with a sale of half the developer's stake at commercial operation, the success-path IRR is 16.0% and the cash multiple 5.0x. The developer's peak cumulative cash at risk is about USD 10.2 million.



The risk-weighted net present value weights each stage's spending by the probability of reaching it and the value at close by the probability of closing, at a discount rate that reflects development risk. For Kasiri, at 25 percent and a 15% chance of close from reconnaissance, it is USD -0.95 million.

The value of the position at each stage, described in Chapter 6, prices entry by a co-developer.

14.2 Why the risk-weighted value is negative

On the base-case assumptions, Kasiri is not worth starting, and the reason is not the one usually given. The usual argument is attrition: a project with a 15 percent chance of closing must earn, when it closes, enough to pay for the five or six that fail. That argument holds only if the project creates value when it succeeds. Kasiri does not, at the developer's 25 percent rate. Even if it were certain to close, the developer's net present value at the start of development would be USD -1.00 million. Development money is returned at close at cost plus a 3 percent premium, years after it was spent, and the developer's share of the equity is worth less than it costs at a 15 percent construction-stage rate, because the equity return is below target.

That changes what the break-even values mean. The development premium that sets the risk-weighted value to zero is about 19% of plant cost, which no lender or buyer would accept. A break-even probability of close does not exist on these assumptions: raising the odds only makes it more likely that a loss-making path is followed to the end. The model reports

"n/a: negative on the success path" for it.

The practical consequence is that the value at close must be fixed before the odds are worth improving. Private hydro development in Africa is done by developers with portfolios, by co-developers who enter later at higher probabilities, with grants that fund the riskiest studies, or by states that prepare and tender sites. Each of these lowers the cost of getting to close; none of them helps a project whose success path loses money.

Our research found no public data on development premiums for African hydro [DE:BENCH]. The 3 percent default is a model assumption.

14.3 What changes the answer

The model tests the levers available to a developer.

Case	Risk-weighted NPV (USDm)	Success-path IRR
Base	-0.95	16.0%
Development discount rate 18% instead of 25%	-1.02	16.0%
Development premium 6% instead of 3%	-0.82	18.6%
Stage odds 10 points higher at every stage	-0.96	16.0%
Grant funds half of the feasibility stage	-0.59	18.4%
Tariff 10% higher	-0.76	20.7%
All four development levers together	-0.05	22.1%

Three lessons follow, and the first runs against common practice. Better stage odds on their own do not help: the risk-weighted value barely moves, for the reason given in section 14.2. A lower development discount rate makes things slightly worse, because the success-path return is below 18 percent as well. What helps is anything that raises the value at close or lowers the cost of getting there: a higher premium, a grant for the feasibility stage, or a higher tariff. Only all the levers together bring the risk-weighted value close to zero. A developer looking at numbers like these should first negotiate the tariff and the premium, then decide how much to spend on improving its odds.

14.4 Value step-ups

Value is created at two moments after development. At financial close, the private equity's value at a construction-stage rate of 15 percent, net of contributions, is about USD -5.1 million: slightly negative, because the equity return is below the 15 percent rate. Between close and commercial operation, construction risk falls away and the rate falls to 11 percent; the step-up in the value of the operating equity is about USD 18.3 million in present-value terms. That step-up is what a developer captures when it sells part of its stake after commercial operation.

14.5 Sell-down

In the model, the developer holds 40 percent of the private equity after close and sells half of that stake at commercial operation at the value of the operating equity. The sale brings in about USD 19.5 million. At Bugoye in Uganda, the original sponsors exited in 2015, according to search summaries we could not confirm in a primary source, and the buyer refinanced the plant in 2017 [DE:S11; DE:S30]; the case illustrates the pattern. Selling at commercial operation recycles capital into the next project; holding captures the long-run yield. The right choice depends on the developer's pipeline and its cost of capital.

14.6 What the Kasiri case shows

Kasiri is attractive to a developer who is already at the financing stage and unattractive to one at reconnaissance. Tamarind's position at the start of the permitting stage is worth about USD 1.28 million on the developer's assumptions, which sets the starting point for the co-developer negotiation in Chapter 18. A modest increase in the tariff, enough to bring the equity return to its target, would raise that value and close the financing gap at the same time.

Points for the investment and credit committees

1. Ask for the developer's return on the success path and weighted by the probability of failure, and for its peak cash at risk.
2. Price a co-developer's entry from the value of the position at the stage of entry, not from the money already spent.
3. Treat the development premium as a model assumption until evidence for the specific market exists.
4. Show the step-up between close and operation, and how the developer intends to capture it.
5. Check that the developer has the portfolio or the partners to absorb failed projects.

Working with the model

01A_DEVELOPMENT holds every developer measure: the stage table, the spend profile, the risk-weighted value, the break-even premium and probability, the value by stage, the step-ups, the sell-down and the developer's cash flow and IRR. The developer sensitivities are in the full-engine snapshot.

Chapter 15. The public side: affordability, support and contingent liabilities

This chapter supports the decision on what support to ask the state for, and whether the state can carry it. It is written for developers and lenders as much as for finance ministries, for a simple reason: a support package the ministry will not sign does not close. The chapter condenses the author's earlier policy paper on hydro and public liabilities into the questions a transaction team must answer.

Place in the analytical chain: public obligations, the link that decides whether the arrangements behind bankability are affordable.

15.1 Can the buyer pay?

A project finance model usually assumes the buyer pays. In Sub-Saharan Africa that assumption needs testing. Transmission and distribution losses averaged about 15 percent of energy dispatched across the region, and about 23 percent excluding South Africa; bill collection ranged from about 58 to close to 100 percent [UT-02]. The power sector's quasi-fiscal deficit averaged about 1.5 percent of GDP [UT-02].

The model builds a simplified cash model of the buyer, translating local-currency flows into dollars. From it, the maximum sustainable PPA payment is the utility's cash available after displaced supply, before paying the new PPA, divided by a coverage buffer of 1.2:

$$MSP_t = \max(0, CASH_t) / 1.2$$

$$GAP_t = \max(0, PPA_t - MSP_t)$$

For Kasiri, the buyer's payment capacity is about 6.0x times the PPA bill in the worst of the first ten years. A 60 MW plant is small in a national system, and the buyer can pay it.

15.2 Can the buyer afford it?

Being able to pay is not the same as being better off. Kasiri's tariff is above what the utility collects for each megawatt-hour it sends out, after losses and non-payment. Every unit it buys from Kasiri therefore costs it money. The model measures that through a counterfactual: the utility's cash with and without the project. The central government's fiscal NPV from Kasiri, counting taxes, royalties and transmission, is about USD 35 million; the consolidated figure for government and state utility together is about USD –28 million at an 8 percent discount rate.

A negative consolidated figure is not a reason to reject the project. The utility may need the energy, the alternatives may cost more, and unserved demand has a cost that the model does not price. It is a reason to know the number before signing, and to compare the tariff with the utility's cost of the alternative.

15.3 Where a shortfall goes

When the buyer cannot pay, the shortfall goes through three doors in order: a budget transfer if the government has committed to one, a sovereign payment guarantee if the PPA has one (capped at a number of months of billing and reimbursed when the utility recovers), and otherwise arrears owed to the project.

With Kasiri's buyer under stress, the order matters more than the size. In the offtaker stress, with collections down 8 points, transfers cut by 30 percent and the retail tariff frozen for five years, the utility can no longer pay at all within ten years. With a budget backstop, the project's lenders see nothing, and the consolidated fiscal NPV falls to about USD

–125 million. Without one, the twelve-month guarantee is exhausted in the first year of the shortfall, the remainder becomes arrears, and the project defaults: its minimum DSCR falls to –0.53x.

15.4 Direct support and contingent liabilities

Direct support is cash the state spends: equity, grants, transmission and budget transfers. Contingent liabilities are payments it may have to make: debt guarantees, PPA guarantees, termination payments and currency undertakings. The model reports the maximum simultaneous exposure, defined as the larger of the termination amount and the sum of guarantees, so that the two are not added.

For Kasiri, the peak contingent exposure is about USD 225 million, about 0.6% of Navaria's fictional GDP, mostly the termination amount. Multilateral guarantees, where used, are backed by indemnities from the government and count as its exposure [A2:S53].

15.5 The fiscal screen

The IMF and World Bank debt sustainability framework for low-income countries and the PPP fiscal risk assessment model are the standard tools for assessing public debt and PPP risks [FR-09; FR-01]. The model's screen does not replace them. It asks a narrower question: does this one project add material fiscal pressure? For Kasiri, the answer is "LOW additional fiscal pressure".

15.6 A contrast: Lumora Falls

The earlier policy paper's 400 MW Lumora scheme gave a different answer. Its contingent exposure at commercial operation was of the order of 3 percent of GDP, its fiscal screen was moderate in the base case and high under cost overrun or offtaker stress, and the offtaker stress turned a small positive central-government fiscal NPV into a loss of more than a billion dollars in present value. Size, a capacity-heavy tariff and a state-built line put Lumora's risks on the public side. Kasiri's size and energy-only tariff keep most of its risks with the developer and the lenders. The method is the same; the answer depends on the project.

Points for the investment and credit committees

1. Test the buyer's payment capacity before negotiating the tariff, and report the ratio for the worst of the first ten years.
2. Compare the tariff with what the utility collects per megawatt-hour, and report the consolidated fiscal effect alongside the Treasury view.
3. Make the routing of a shortfall explicit: budget, capped guarantee or arrears.
4. Record every guarantee and termination obligation, with its counter-indemnity, as state exposure.
5. Ask for the ministry's written sign-off on the support package and the fiscal screen before close.

Working with the model

12_UTILITY holds the utility cash model, the maximum sustainable payment and the no-project counterfactual. 16_REVENUE routes the shortfall. 22_GOVERNMENT_SUPPORT to 26_DEBT_SUSTAINABILITY hold direct support, guarantees, contingent liabilities, fiscal impact and the screen.

Chapter 16. Risk allocation and stress testing

This chapter supports the decision every lender and investor makes before committing: what breaks first, when, and who holds it. Stress testing is often presented as a list of sensitivities. It is more useful as a search for the point at which the project fails.

Place in the analytical chain: stress, which brings the earlier links together for the people who carry the risk.

16.1 Who carries what

The model maps sixteen risks against eight parties: government, developer, construction contractor, lender, utility, consumer, insurer and development finance institution. The allocation for a private IPP with an energy-only tariff differs from the earlier paper's PPP in one important way: the developer is the primary bearer of hydrology risk, because the tariff pays only for energy delivered.

Risk	Primary bearer at Kasiri	Main instruments	Model stress
Hydrology	Developer	P90 sizing, debt service reserve, cash sweep	Drought, climate, P90 cash flows
Construction and overrun	Contractor and developer, by contract structure	Contract structure, contingency, standby equity	Overrun, delay
Transmission and grid	Developer (own interconnector); utility (grid)	Construction contract, deemed energy for grid curtailment	Transmission delay
Offtaker payment	Utility, then government	Payment security, guarantee, liquidity facility	Offtaker stress, FX step
Currency	Utility and government	Partial local-currency tariff, convertibility undertaking	FX step
Interest rate	Developer, for the unhedged share	Hedging	Interest +300 bp
Political and termination	Government	Political risk insurance, termination compensation	Termination exposure

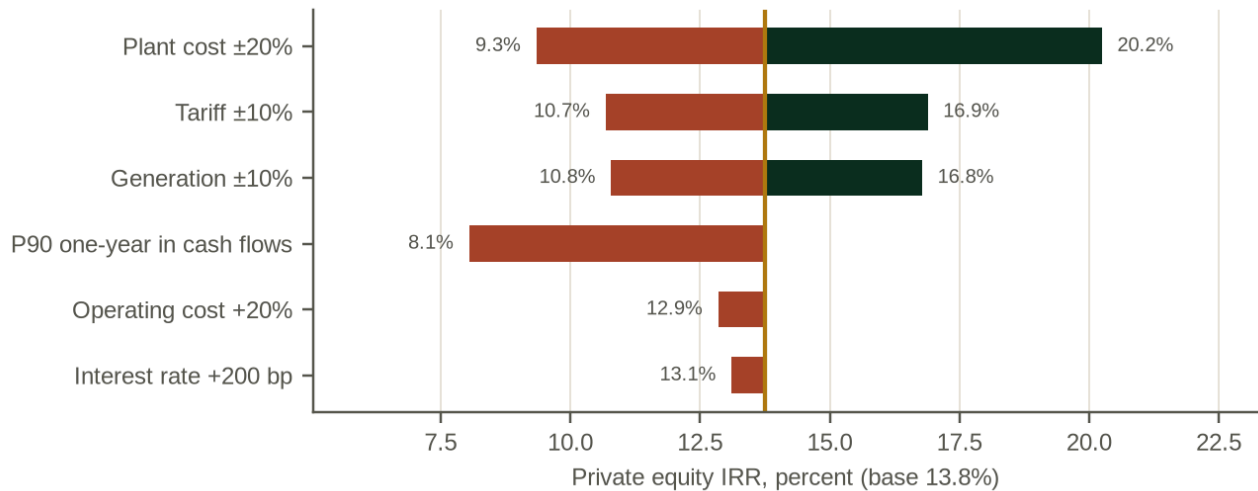
16.2 Stress results

All stresses below hold the financing package at its base-case terms.

Table 16.1. Kasiri River Hydro: stress results with the financing package locked

Scenario	Equity IRR	Minimum DSCR	Financing gap (USDm)	Utility payment capacity	Consolidated fiscal NPV (USDm)
Base	13.8%	1.53x	4.1	6.0x	-28
Low case	9.1%	1.36x	16.2	6.0x	-106
High case	16.6%	1.70x	3.9	5.8x	-26
Three-year drought	11.0%	0.69x	4.1	6.0x	-27
P90 one-year in every year	8.1%	1.16x	4.1	7.2x	-29
Climate trend	12.8%	1.46x	4.1	6.0x	-29
Overrun 27%	10.2%	1.52x	22.1	6.0x	-30
Overrun 96%	5.7%	1.49x	68.3	6.0x	-37
Two-year delay	10.1%	1.55x	23.8	6.8x	-22
Transmission two years late	13.8%	1.31x	3.9	5.3x	-61
Interest +300 bp (unhedged share)	12.8%	1.43x	6.2	6.0x	-28
Offtaker stress, budget backstop	13.8%	1.53x	4.1	0.0x	-125
Offtaker stress, no backstop	-100.0%	-0.53x	4.1	0.0x	-152
FX step 50%	13.8%	1.53x	4.1	0.5x	-111
Combined (overrun, delay, offtaker, FX)	7.8%	1.54x	43.1	0.0x	-157

Note: An equity IRR of -100% means the equity is wiped out. Utility payment capacity is the worst ratio of the first ten operating years.



16.3 What breaks first

Three results stand out.

The river breaks the debt first. Under an energy-only tariff, a three-year drought at 55 percent of normal flow pushes the minimum DSCR to 0.69x. The reserve is drawn but does not cover the whole shortfall. A lender sizing on the ten-year P90 needs either a larger reserve, a cash sweep in wet years, or completion of a drought test before close.

The buyer breaks the project only if the state steps aside. With a budget backstop, the offtaker stress costs the state and leaves the project whole; without one, the project defaults. The project's credit is therefore partly the state's credit, whatever the structure says.

Construction overruns break the equity, not the debt. With the financing locked, an overrun at the reference-class mean leaves a financing gap of about USD 68 million that only standby equity or new money can fill.

16.4 Reverse stress

Reverse stress asks how far a variable must move before the project fails. With the financing locked, the minimum DSCR reaches 1.0 when river flows fall by about 28% in every year. That is a larger move than the one-year P90 implies on average, and a smaller one than a severe multi-year drought produces in the years it lasts, which is why the drought test matters more than the average.

Points for the investment and credit committees

1. Test with the financing package locked, and read the financing gap as well as the DSCR.
2. Test at least one multi-year drought under an energy-only tariff, and check the reserve against it.
3. Run the offtaker stress with and without a budget backstop, and record who carries the difference.
4. Ask for reverse stress results: the flow reduction and the overrun at which the project fails.
5. Read the allocation matrix for the specific tariff and structure; do not borrow it from another project.

Working with the model

27_SCENARIOS holds the cases, stress parameters and the full-engine scenario snapshot; 28_SENSITIVITY holds the tornado and the sensitivity snapshot. 29_RISK_ALLOCATION holds the matrix. Stress toggles are on 01_CONTROL_PANEL.

Chapter 17. Financial close: conditions precedent, readiness gates and the close

This chapter supports the final decision of development: go, conditional go or stop for financial close, and on what evidence. Financial close is not a signing ceremony. It is the moment when every condition the lenders set has been met and the first money can be drawn, and many projects that "signed" never get there.

Place in the analytical chain: financial close, the end of development and the start of construction.

17.1 What financial close means

Three moments are often confused. Signing is when the financing agreements are executed. Satisfaction of conditions precedent is when every condition in those agreements has been met, certified and accepted by the lenders. First drawdown is when money moves. Financial close, in the sense that matters, is the second: the lenders are obliged to fund, subject only to ongoing conditions. A project can sign and then spend months satisfying conditions, and some never do.

17.2 Conditions precedent by workstream

Lenders' conditions precedent fall into seven workstreams. Annex F gives a full checklist; the essentials are below.

Workstream	Typical conditions
Technical	Lenders' technical adviser's report on design, energy yield, construction contracts, budget and schedule
Hydrology	Independent energy assessment, flow record review, P50 and P90 agreed
Environmental and social	Lenders' E&S adviser's report; environmental and social action plan agreed; resettlement on schedule
Legal	All project contracts signed and effective; permits held; direct agreements; security perfected; legal opinions
Commercial	PPA and implementation agreement effective; payment security in place; tariff approved
Financial	Financial model audited; equity committed; hedging in place; insurance placed; accounts opened
Public sector	Government support approved; guarantees and indemnities executed; exchange control approvals

17.3 The readiness gates

The model turns those conditions into 23 gates, each marked critical or not, each counted only on evidence. Model tests are computed automatically; other gates are counted only when the evidence status is entered as met. The decision follows a fixed rule: STOP if a critical gate is not met or has no evidence, NOT READY if critical gates are only partly met, CONDITIONAL GO if every critical gate is met, and GO only when all 23 are. A GO says the evidence file is complete for lenders and sponsors to decide; it is not a recommendation.

Table 17.1. Kasiri River Hydro: financial close readiness gates

#	Gate	Area	Critical	Status
1	Flow record of at least 15 years and independent hydrology review	Resource	Yes	NOT MET (model test)
2	P90 energy confirmed by the lenders' technical adviser	Resource	Yes	PARTIAL
3	Bankable feasibility study signed off by the lenders' technical adviser	Design	Yes	NOT MET (model test)
4	Geotechnical investigation sufficient for a baseline report	Design	Yes	PARTIAL
5	ESIA approved and compliant with lender standards	E&S	Yes	NOT MET (model test)
6	Resettlement action plan approved and funded	E&S	Yes	MET (model test)
7	Generation licence and water-use permit granted	Permits	Yes	PARTIAL
8	Land rights secured for all project areas	Permits	Yes	NOT MET
9	PPA signed and approved by the regulator	Revenue	Yes	PARTIAL
10	Implementation or concession agreement signed	Revenue	Yes	PARTIAL
11	Grid connection agreement signed and transmission financed	Grid	Yes	MET (model test)
12	Transmission in service no later than plant COD	Grid	No	MET (model test)
13	EPC contract(s) signed with fixed price, completion date and LDs	Construction	Yes	PARTIAL
14	O&M arrangements and owner's team in place	Operations	No	PARTIAL
15	Payment security of at least 6 months in place	Offtaker	Yes	NOT MET (model test)
16	Offtaker payment capacity at least 1.2x the PPA bill (worst of first 10 years)	Offtaker	Yes	MET (model test)
17	Financing plan fully committed (no financing gap)	Finance	Yes	NOT MET (model test)
18	Minimum DSCR at or above the sizing target in the base case	Finance	Yes	MET (model test)
19	Equity commitments signed and equity IRR at or above target	Finance	Yes	NOT MET (model test)
20	Political risk cover or guarantees signed	Risk	No	NO EVIDENCE
21	Government support approved by the finance ministry; fiscal screen not HIGH	Public finance	Yes	MET (model test)
22	Insurance programme placed (construction all risks, DSU)	Risk	No	PARTIAL
23	Independent model audit completed	Finance	No	NO EVIDENCE

Note: Statuses marked as model tests are computed by the model; the others are evidence statuses entered for the case.

Kasiri meets 6 of the 23 gates. The decision reads "STOP: a critical gate is not met". Seven critical gates are not met: the flow record and its independent review, the bankable feasibility sign-off, an ESIA approved to lender standards, land rights, payment security of six months, a fully committed financing plan, and an equity return at target. Several others are partly met. None of them is a surprise for a project at Kasiri's stage, and each points to a specific task.

17.4 Funds flow at close

At close, the sources and uses table becomes a funds flow: who pays what into which account on which day. Development costs are reimbursed to the developer within the agreed cap, the development premium is paid, fees are paid, the debt service reserve is funded, and the first construction payments are made. The model's sources and uses on 17_PROJECT_FINANCE are the starting point; a funds flow memorandum agreed by all parties replaces it.

17.5 The investment memo

The decision to close is recorded in two memos: the sponsor's investment committee paper and the lender's credit paper. Annex H gives a template. Both should place the readiness decision next to the recommendation, list the open gates as conditions, and show the project's results on the success path, under stress and on the public side.

Points for the investment and credit committees

1. Distinguish signing, satisfaction of conditions precedent and first drawdown, and plan for the gap between them.
2. Track conditions precedent by workstream, with an owner and a date for each.
3. Count a gate as met only on evidence, with the document, the signatory and the date recorded.
4. Treat a model test that fails as a finding about the project, not a gap in the file.
5. Record the readiness decision next to the recommendation in every committee paper.

Working with the model

30A_CLOSE_READINESS holds the 23 gates, their evidence status, the automatic tests and the decision. *30_BANKABILITY* holds the nine-gate screening framework used earlier in development. *17_PROJECT_FINANCE* holds sources and uses.

Chapter 18. The Kasiri case from site to close

This chapter supports two decisions taken together, as they usually are in practice: whether a co-developer should fund the rest of Kasiri's development, and on what terms, and whether a group of development finance lenders should start due diligence. It walks the case through the analytical chain of the book and ends with a recommendation and the conditions for moving the readiness decision from STOP to CONDITIONAL GO.

Place in the analytical chain: the whole chain, applied to one project.

18.1 The ask

Tamarind Hydro has completed reconnaissance, pre-feasibility and a feasibility study for Kasiri River Hydro, a 60 MW run-of-river plant in fictional Navaria. It has spent about USD 5.2 million of its own money. Two requests are on the table.

The first goes to a co-developer: fund the remaining USD 3.8 million of development, which covers permitting, the negotiation of the PPA and the financing stage, in exchange for a share of the development company. The second goes to a group of development finance institutions: issue a mandate to arrange senior debt of about USD 145 million over 16 years, on the basis of the feasibility study and the model.

The two requests depend on each other. The co-developer's money buys a position whose value rests on the lenders' terms, and the lenders will not commit until the work that the co-developer's money pays for has been done.

18.2 The project in one table

Table 18.1. Kasiri River Hydro: base case at a glance

Item	Value
Installed capacity	60 MW
Mean generation (P50)	292 GWh a year; capacity factor 56%
P90, one year and ten-year average	236 and 268 GWh
Plant cost (2026 dollars)	USD 157 million; USD 2,614 per kW
Total uses of funds at close	USD 215 million
Senior debt and gearing	USD 145 million; 68%
Tariff	USD 112 per MWh, energy only, half indexed to US inflation
Levelised cost of energy	USD 106 per MWh
Private equity IRR	13.8% against a 15.0% target
Minimum DSCR, P50 case	1.53x
Financing gap	USD 4.1 million
Developer IRR on the success path	16.0%
Fiscal screen	LOW additional fiscal pressure
Readiness decision	STOP: a critical gate is not met; 6 of 23 gates met

The table already tells most of the story. The plant is a sound one, with a good capacity factor, a moderate cost for its size and a buyer able to pay. It is slightly short of the tariff its sponsors need, and it is far from ready for close.

18.3 The development question

What is the co-developer buying? Not the USD 5.2 million that Tamarind has spent: that money is sunk, and a buyer pays for what the position is worth from here. At the start of the permitting stage, the model gives a probability of reaching close of 58%. Weighted by that probability and discounted at 25 percent, the value received at close is worth about USD 3.34 million today, and the remaining spend about USD 2.06 million. The difference, USD 1.28 million, is the value of the whole position before the co-developer's money comes in.

That gives a simple rule for the co-developer's share. If it funds all the remaining spend, a fair share of the value at close is the ratio of the remaining spend to the value at close, both risk-weighted: about 62%. Tamarind keeps the rest, which is worth the value of its position. A co-developer with a lower discount rate, or with a better view of the odds, would accept a smaller share; one with a higher rate would ask for more. The rule treats the reimbursement of all development costs at close, including Tamarind's USD 5.2 million, as value shared between the two. That is why the position is worth something at this stage although the project as a whole loses money from reconnaissance at 25 percent (Chapter 14): the sunk costs are recovered from the financing if the project closes. A term sheet that pays each party's reimbursement to the party that funded it would change the split, and should be modelled before the share is agreed.

Three refinements matter in the negotiation.

The money should be committed in tranches, one per stage, each released when the previous stage succeeds. The co-developer then never funds a stage that the project did not reach, and its risk-weighted cost falls.

The share should be adjusted at close for the development premium and the reimbursement of costs that the lenders actually allow, because both are paid out of the financing and both are uncertain until the term sheet is signed.

The tariff is the largest single driver of value. A tariff that brings the equity return to its target would raise the value of the position at every stage, as Chapter 14 showed. A co-developer whose share is fixed now captures part of any later improvement in the tariff; Tamarind should either negotiate the tariff first or agree a share that steps down if the tariff improves.

18.4 The credit question

What would lenders see in due diligence? On the base case, a project that meets the usual tests with a margin: a minimum DSCR of 1.53x on the P50 case and an LLCR of 1.59x, with debt sized on the ten-year P90 and the 70 percent gearing limit binding by a small margin.

Under stress, three findings would come out of the lenders' model review, and they should come out of the developer's first.

The drought test fails. A three-year drought at 55 percent of normal flow takes the minimum DSCR to 0.69x. The debt service reserve is not enough on its own. The lenders will ask for a larger reserve, a cash sweep in wet years, a smaller loan, or a combination.

The overrun test leaves a gap that no one has committed to fill. With the financing locked, an overrun at the reference-class mean leaves about USD 68 million unfunded. The lenders will ask for standby equity or a contingent loan, sized against the overrun they believe, and for a contract structure that moves part of the risk to the contractor.

The buyer is sound in the base case but not under stress. With collections down, transfers cut and the retail tariff frozen, the utility can no longer pay within ten years. With a budget backstop the project is protected; without one, it defaults. The lenders will ask for six months of payment security, a liquidity facility, or a government payment undertaking, and the ministry will have to decide whether it signs.

None of these findings is unusual. Each is a condition that can be met before close, and each has a cost that should be in the model before the mandate is signed.

18.5 The public question

What does Navaria take on? Its direct support is nil: the developer builds the line and the state contributes no equity or grant. Its contingent exposure is the termination amount under the implementation agreement, which peaks at about USD 225 million, about 0.6% of GDP, and any guarantee it gives the buyer. The central government gains about USD 35 million in present value from taxes and royalties. The consolidated figure for government and utility together is about USD -28 million, because the utility pays more for Kasiri's energy than it collects from its customers.

That last number is the one the ministry will look at if the developer asks for a tariff increase. A higher tariff closes the developer's gap and widens the utility's. The case for it rests on what the utility would otherwise pay for the energy, or on the cost of the demand it would otherwise not serve, and the developer should bring that analysis to the table rather than wait to be asked.

18.6 Readiness

The readiness sheet reads "STOP: a critical gate is not met", with 6 of 23 gates met. That is the right answer for a project at the start of permitting, and it is not a reason to stop developing. It is a reason not to commit to close, and it sets the work programme.

Table 18.2. Work programme from STOP to CONDITIONAL GO

Critical gate not met	Task	Stage	Funded by
Flow record and independent review	Install and calibrate gauges; independent energy assessment	Permitting	Co-developer tranche 1
Bankable feasibility sign-off	Lenders' technical adviser reviews design, cost and schedule	Financing	Co-developer tranche 3
ESIA to lender standards	Gap analysis against IFC Performance Standards; action plan	Permitting	Co-developer tranche 1
Land rights	Acquisition and compensation plan; titles or long leases	Permitting	Co-developer tranche 1
Payment security of six months	Letter of credit or liquidity facility in the PPA	PPA	Co-developer tranche 2
Financing plan fully committed	Close the equity gap through the tariff, lower costs or more committed equity	PPA and financing	Co-developer tranches 2 and 3
Equity return at target	Tariff, indexation or deemed-energy terms; or lower costs	PPA	Co-developer tranche 2

A project that completes this programme reaches CONDITIONAL GO when every critical gate is met, and GO only when the remaining non-critical gates are too.

18.7 Recommendation

To the co-developer: proceed, on three conditions. Commit the USD 3.8 million in three tranches, one each for permitting, the PPA and financing, each released on the success of the previous stage. Price the entry from the value of the position at the start of permitting, not from Tamarind's sunk costs, which implies a share of the value at close of about 62% at a 25 percent discount rate. Make the share adjustable for the tariff achieved in the PPA and for the development premium and reimbursement the lenders allow.

To the lenders: do not issue a mandate yet. Agree an indicative term sheet that sets the generation case for sizing, the drought test, the minimum reserve and the standby equity requirement, so that the developer and the co-developer can price them now. Revisit at the end of the PPA stage, when the payment security and the tariff are known.

To the developer: settle the tariff question before the co-developer's share is fixed. On the model, the tariff that gives the target return in the IPP structure is about USD 116 per MWh, against 112. Bring an independent view of the utility's avoided cost to the negotiation, and test a drought reserve and a standby equity line in the model before the term sheet is discussed.

18.8 What would change the recommendation

The recommendation would change if the flow record under way showed a mean flow materially below the feasibility study's estimate; if the utility's finances deteriorated before the PPA was signed; if the construction contract came in above the reference-class mean; or if the ministry declined to provide any payment support. Each of these would lower the value of the position and the co-developer should be free to stop at the next tranche. It would also change, in the other direction, if a grant were found for the permitting studies or the tariff were agreed at the level the sponsors need.

Points for the investment and credit committees

1. Price a development position from its value at the stage of entry, not from the money already spent.
2. Commit development money in tranches tied to stage success.
3. Fix the share only after the tariff, or make it adjustable for the tariff achieved.
4. Bring the lenders' stress tests forward into development, so that their cost is known before the mandate.
5. Read STOP as a work programme, not as a verdict on the project.

Working with the model

Every number in this chapter comes from the base case of MODEL 7 and the full-engine snapshot. *01A_DEVELOPMENT* section D2 gives the value of the position by stage; *30A_CLOSE_READINESS* gives the gates; *27_SCENARIOS* gives the stress results; *32_DASHBOARD* gathers the headline results. To test the recommendation, change the tariff, the stage probabilities or the developer's discount rate on *01_CONTROL_PANEL* and *01A_DEVELOPMENT* and read the value of the position again.

Annexes

The annexes are working tools. Annexes A to I are checklists and templates for use on a live transaction; Annex J lists the public benchmark cases; Annex K records how the model was tested and what it does not do; Annex L lists every source cited, with its verification status; Annex M is a glossary.

Annex A. Key results: definitions

Result	Definition	Where in MODEL 7
P50, P90 one-year, P90 ten-year	Mean annual generation; generation exceeded in nine years out of ten; ten-year average exceeded with 90 percent probability	03_HYDROLOGY
Capacity factor	P50 generation divided by installed capacity times 8,760 hours	03_HYDROLOGY
Total uses	Plant cost, project-built transmission, development costs and premium, interest during construction, fees and the initial debt service reserve, all nominal	17_PROJECT_FINANCE
Gearing	Senior debt divided by total uses	17_PROJECT_FINANCE
CFADS	Cash flow available for debt service: revenue less operating costs, taxes and maintenance capital spending	20_CASH_FLOW
DSCR	CFADS divided by scheduled debt service in a year; minimum and average over the loan life	20_CASH_FLOW
LLCR	Present value of CFADS over the loan life divided by debt outstanding at commercial operation	17_PROJECT_FINANCE
Project IRR	Post-tax, nominal, on all capital	17_PROJECT_FINANCE
Private equity IRR	Nominal, on private equity contributions and distributions	17_PROJECT_FINANCE
Financing gap	Private equity required less private equity committed	17_PROJECT_FINANCE
LCOE	Levelised cost of energy delivered, nominal	17_PROJECT_FINANCE
Developer IRR	Success path, from first study to last distribution, with sell-down	01A_DEVELOPMENT
Risk-weighted developer NPV	Probability of close times the present value of value at close, less risk-weighted development spend, at the development discount rate	01A_DEVELOPMENT
Value of the position by stage	Risk-weighted value at close less risk-weighted remaining spend, at the start of each stage	01A_DEVELOPMENT
Utility payment capacity	Maximum sustainable PPA payment divided by the PPA bill, worst of first ten operating years	12_UTILITY
Fiscal NPV, central and consolidated	Present value of net flows to central government, and to government and state utility together, at 8 percent	25_FISCAL_IMPACT
Peak contingent exposure	Larger of the termination amount and the sum of guarantees, at its highest	24_CONTINGENT_LIABILITIES
Readiness decision	STOP, NOT READY, CONDITIONAL GO or GO on 23 gates	30A_CLOSE_READINESS

Annex B. Development budget template

Stage	Workstream	Activity	Budget (USD '000)	Months	At risk or reimbursable	Funded by	Exit test
1. Reconnaissance	Technical	Desk study, site visit, preliminary head and flow			At risk		Site worth a pre-feasibility study
2. Pre-feasibility	Technical	Layout options, cost estimate, energy estimate			At risk		Least-cost option clears a tariff test
2. Pre-feasibility	Hydrology	Gauge installation and calibration begins			At risk		Record running
3. Feasibility	Technical	Bankable design, geotechnical campaign, cost and schedule			Reimbursable within cap		Feasibility accepted by the developer
3. Feasibility	E&S	ESIA and resettlement planning to lender standards			Reimbursable within cap		ESIA submitted
4. Permitting	Legal	Licence, water-use permit, land acquisition			Reimbursable within cap		Permits held
5. PPA	Commercial	PPA, implementation agreement, grid connection			Reimbursable within cap		PPA initialled
6. Financing	Finance	Lender advisers, model audit, insurance, legal			Reimbursable within cap		Close
All	Management	Developer's team, travel, overheads			At risk		

Record for each line the probability of moving to the next stage, the latest date the spend can be deferred to, and the cap on reimbursement in the term sheet.

Annex C. Consents tracker

Consent	Issuing authority	Law or regulation	Pre-condition	Applied	Expected	Granted	Expiry	Conditions	Owner
Permit to study									
Water-use permit									
Environmental approval			ESIA						
Generation licence									
Land rights			Valuation and compensation						
Construction permit									
Grid connection approval			Grid study						
Tariff approval			PPA						
Exchange control approval			Financing terms						
Tax incentives			Investment certificate						

Annex D. Construction contract term sheet checklist

1. Structure: single EPC, split civil and electromechanical, or multi-contract, and who carries interface risk.
2. Price: lump sum, with the items excluded and the provisional sums listed; price adjustment formula, if any.
3. Ground risk: geotechnical baseline report, who carries conditions outside it, and how claims are measured.
4. Time: completion date, milestones, extension of time events and the procedure for claiming them.
5. Delay liquidated damages: rate per day, cap, and comparison with debt service and fixed costs during delay.
6. Performance: output and efficiency guarantees, test procedure, performance liquidated damages and cap.
7. Security: performance bond, advance payment bond, retention, parent company guarantee.
8. Overall cap on liability and the exclusions.
9. Defects liability period and warranties on equipment.

10. Insurance: construction all risks, delay in start-up, who places it.
11. Change and variation procedure; owner's engineer's role.
12. Dispute resolution: dispute board, arbitration seat and rules.
13. Termination rights and payments on each side.
14. Lenders' direct agreement and step-in rights.
15. Local content and labour conditions, E&S obligations flowing from the ESIA.

Annex E. PPA checklist

1. Term, start date, and conditions for commercial operation.
2. Tariff: energy and capacity components, currency, indexation, and how each is billed.
3. Deemed energy: for grid unavailability and curtailment, how it is measured, and any annual cap.
4. Hydrology risk: who carries it, and whether there is any minimum take or banking of energy.
5. Payment: invoicing, due date, late-payment interest, and the order of application of payments.
6. Payment security: letter of credit or escrow, number of months, replenishment, and any government or liquidity support behind it.
7. Change in law and change in tax, and the compensation mechanism.
8. Force majeure: natural and political, relief and payments during each.
9. Termination events on each side, and the termination payment formula for each, distinguishing debt and equity.
10. Transfer of the plant at the end of the term, if any.
11. Dispute resolution and governing law.
12. Lenders' direct agreement, assignment and step-in.
13. Dispatch and grid code obligations, outage planning.
14. Metering, testing and data.
15. Regulator's approval of the tariff and of the PPA itself.

Annex F. Conditions precedent checklist

Workstream	Condition	Evidence	Owner	Status
Technical	Lenders' technical adviser's final report	Report and reliance letter		
Hydrology	Independent energy assessment with P50 and P90	Report		
Hydrology	Flow record reviewed and accepted	Adviser's letter		
E&S	Lenders' E&S adviser's report; action plan agreed	Report; signed action plan		
E&S	Resettlement action plan approved and funded	Approval; funding evidence		
Legal	All project agreements signed and effective	Certified copies; effectiveness certificates		
Legal	Permits and licences held	Certified copies		
Legal	Land rights secured	Titles or leases		
Legal	Security perfected; direct agreements signed	Registrations; signed agreements		
Legal	Legal opinions	Opinions from lenders' and borrower's counsel		
Commercial	PPA and implementation agreement effective	Effectiveness certificates		
Commercial	Payment security issued	Letter of credit or facility agreement		
Commercial	Tariff approved	Regulator's decision		
Financial	Financial model audited	Auditor's report		
Financial	Equity committed; standby equity in place	Equity contribution agreement; guarantees		
Financial	Hedging executed	Confirmations		
Financial	Insurance placed	Broker's letter; certificates		
Financial	Accounts opened	Account bank confirmations		
Public sector	Government support approved	Ministry of finance approval		
Public sector	Guarantees and indemnities executed	Signed instruments		
Public sector	Exchange control approvals	Central bank letter		

Annex G. Lender data request

1. Flow record: daily or monthly flows, gauge history, rating curves, gaps and how they were filled, and any regional correlation used to extend the record.
2. Energy model: inputs, losses, environmental flow, outages, and the P50 and P90 calculation.
3. Feasibility study: design, geotechnical data, quantities, cost estimate and schedule.
4. Construction contracts: drafts or signed versions, bids received, and the evaluation.
5. Financial model: the model, the assumptions book and the model audit scope.
6. Offtaker: last five years of audited accounts, tariff decisions, collection rates, losses and arrears to existing IPPs.
7. Permits and land: copies, conditions and status.
8. E&S: ESIA, resettlement action plan, stakeholder engagement record and grievance log.
9. Government support: drafts of the implementation agreement, guarantees and any support letters.
10. Sponsors: corporate structure, accounts, track record and equity sources.
11. Insurance: broker's programme and indications.
12. Transmission: grid study, connection agreement, line design and cost.

Annex H. Investment memo template

1. Recommendation, in one paragraph, and the readiness decision next to it.
2. The project in one table (as Table 18.1).
3. The development position: spend to date, remaining spend, value of the position by stage, terms of entry.
4. The commercial structure: PPA, implementation agreement, payment security, government support.
5. Construction: contract structure, cost, contingency, schedule, overrun and delay protection.
6. Financing: sources and uses, lender group, debt terms, binding sizing constraint, reserves.
7. Returns: project and equity IRR on the base case; developer IRR and risk-weighted value.
8. Stress results with the financing package locked, including drought, overrun at the reference-class mean, and off-taker stress with and without a backstop.
9. The public side: utility payment capacity, consolidated fiscal NPV, contingent exposure and fiscal screen.
10. Open gates and conditions, each with owner and date.
11. What would change the recommendation.

Annex I. Government support note

A short note for the ministry of finance, to accompany any request for support.

1. The support requested: equity, grant, transmission, budget transfer, payment guarantee, termination payment, currency undertaking.
2. Why it is needed: the specific risk it covers and why the market will not carry it.
3. Its size: maximum simultaneous exposure, its profile over time, and its share of GDP and revenue.
4. The routing of a utility shortfall: budget, capped guarantee or arrears, and the cap.
5. The consolidated fiscal effect, for government and utility together, with and without the project.
6. The counter-indemnities given to any multilateral guarantor.
7. The fiscal screen result and the stress case that would change it.
8. The disclosure the government will make of the commitment.

Annex J. Public benchmark cases

Table J.1. Public benchmark cases (as published; nominal, unadjusted)

Project	Country	Structure	MW	USD/kW	Timeline	Credit support
Nachtigal	Cameroon	IPP/PPP, IFC-led DFI debt	420	3,293	FC Nov 2018; plan 5 yrs; full COD slipped to Dec 2024 / 2025	IBRD payment guarantee EUR 86m + loan guarantee EUR 171m; MIGA; 75:25 debt:equity
Bujagali	Uganda	IPP + IDA PRG	250	3,448	c.2007-2012	IDA PRG US\$115m with sovereign indemnity; MIGA US\$115m; liquidity facility
Bui	Ghana	Public (BPA), China Exim, cocoa escrow	400	1,975	COD Dec 2013	China Exim tranches secured on cocoa export escrow
Kafue Gorge Lower	Zambia	State-owned, sovereign-guaranteed Chinese debt	750	2,667	Nov 2015 → full COD Mar 2023	Sovereign guarantee; ZESCO arrears entered sovereign DSA
Isimba	Uganda	Public, 85% China Exim / 15% GoU	183	3,102	Apr 2015 → Mar 2019	Sovereign borrower
Karuma	Uganda	Public, 85% China Exim / 15% GoU	600	2,833	Contract 2013 (5 yrs) → COD Jun 2024	Sovereign borrower
Julius Nyerere (Rufiji)	Tanzania	100% state budget	2,115	1,348	Construction 2019 → 2025/26 (42-month EPC)	None (budget-funded)
Rusumo Falls	Rwanda/Tanzania/Burundi	Regional public, IDA-financed	80	5,861	Approval 2013 → COD Feb 2025	IDA grants/credits; transmission parallel-financed AfDB/EU
Mpatamanga	Malawi	PPP, storage + peaking	358	4,563	Pre-construction; 5-yr build in 35-yr PPA	IDA PRG US\$100m; MIGA US\$180m; IDA grant US\$350m to govt
Ruzizi III	DRC/Rwanda/Burundi	Regional PPP (stalled)	206	3,689	Target 2030 before conflict pause	IFIs ~60% expected; EIB lead arranger
GERD	Ethiopia	State, domestically financed	5,150	971	2011 → inauguration Sep 2025	Domestic bonds/budget; no MDB finance
Gibe III	Ethiopia	State, ICBC-financed EPC	1,870	n/a	Inaugurated Dec 2016	ICBC loan after MDBs withdrew
Nam Theun 2 (intl.)	Lao PDR	Export IPP, MDB-guaranteed	1,070	1,355	FC Jun 2005 → COD Apr 2010 (search summary)	IDA & ADB PRGs, MIGA (~US\$186m debt covered); EGAT take-or-pay PPA
Upper Trishuli-1 (intl.)	Nepal	DFI-led domestic IPP	216	2,997	Completion slipped 2024 → Dec 2026 (search summary)	DFI debt ~70%
Reventazón (intl.)	Costa Rica	Utility trust & lease, project bond	306	4,583	FC Jan 2014 → inaugurated Sep 2016 (search summary)	Trust owns plant, leased to ICE; IDB A-loan, IFC, IG-rated bond, local-currency bank debt

Note: Sources for each row are in the research case files. Costs reported in euros are not converted (n/a). PUBLIC DATA NOT FOUND means the research found no public figure.

Annex K. Validation summary

How the model was tested

MODEL 7 is built from a single specification. Every formula refers to named inputs and rows, a cycle detector confirms that the model has no circular references, and 14 integrity checks on 33_CHECKS test sources against uses, the cash waterfall, debt balances, reserve balances and the treatment of project-built transmission. The checks confirm internal consistency, not the realism of the assumptions. Every number in this book was produced by recalculating the model and running the scenarios through the full engine.

The adversarial review

Before this edition, the model behind the earlier policy paper was reviewed by an adversarial reviewer who traced formulas and re-ran the model on separate copies. The review reported thirty findings: three critical, ten high, ten medium and seven low. The workbook passed all its own checks while several of these defects were present. The main corrections were:

- the sovereign payment guarantee is capped and reimbursed, so a shortfall beyond the cap becomes arrears to the project;
- the utility can only buy the supply that exists, and unserved demand is shown;
- the fiscal perimeter includes the state utility, through a no-project counterfactual;
- the debt service reserve is drawn first, released only above target and topped up only from available cash;
- locked financing holds every tranche and the commercial principal schedule at base-case values;
- interest during construction and fees are funded within the gearing limit, without circularity;
- the overrun stress at the reference-class mean and a deeper drought were added.

What changed in this edition

This edition added the development module, the construction contracting module and the readiness sheet, an energy-only tariff option, a ten-year P90 lender case with an explicit multi-year drought test, and the Kasiri case. It also corrected the risk-weighted developer value so that it is discounted to the start of development, and replaced a developer IRR that could return a spurious root.

What the model does not do

- It is annual. It has no monthly construction draws, no seasonal dispatch and no peak or off-peak pricing.
- Debt and equity are drawn pro rata with spending, not equity first, which slightly flatters the equity return.
- There is no developer promote or carried interest, and no shareholder loans.
- There is no refinancing; the step-up in value after commercial operation is valued directly.
- The utility model has no balance sheet.
- Stresses are deterministic. There is no Monte Carlo over hydrology and currency and no joint distribution of drought, currency and utility distress.
- Stage probabilities, the development premium and the developer's discount rate are user judgements; no public data were found to calibrate them for African hydro.
- The model has been recalculated in LibreOffice. A test in Microsoft Excel is outstanding.

Annex L. Sources and verification status

Identifiers with two letters and a number (such as HY-08) refer to the research source database; identifiers with a prefix and S-number (such as DE:S1) refer to the numbered source lists of the research files: A1 and A2, African case files; INT, international benchmarks; DE, development evidence. Verification status records what the research team saw. Claims resting on a search summary only should be checked against the original before reliance.

ID	Source	Verification
CL-01	IHA with EBRD and World Bank Group (KGGTF) (2019). Hydropower Sector Climate Resilience Guide. Technical guide (launch page). https://www.hydropower.org/news/new-guide-to-help-hydropower-build-resilience-to-climate-change	Landing or summary page read
CL-03	Conway, Dalin, Landman, Osborn (Nature Energy 2:946-953) (2017). Hydropower plans in eastern and southern Africa increase risk of concurrent climate-related electricity supply disruption. Peer-reviewed article. https://www.lse.ac.uk/granthaminstitute/publication/hydropower-plans-eastern-southern-africa-increase-risk-concurrent-climate-related-electricity-supply-disruption/	Search summary only
CL-04	The Energy Industry Times (news) (2024). Kariba Dam water allocation for electricity generation cut in half. News (secondary). https://teitimes.com/post/kariba-dam-water-allocation-for-electricity-generation-cut-in-half	Search summary only
DE:BENCH	Benchmark table for development model defaults, development evidence file (research note). Records where no public data were found.	Research note
DE:S1	IFC / World Bank (2015), <i>Hydroelectric Power: A Guide for Developers and Investors</i> (Fichtner). https://documents1.worldbank.org/curated/en/917841468188335073/pdf/99392-WP-Box393199B-PUBLIC-Hydropower-Report.pdf	Opened; see research file
DE:S2	L. Shah, IFC InfraVentures, "Early Stage Risk Capital for Renewable Energy", Asia Clean Energy Forum, June 2015. https://asiacleanenergyforum.adb.org/wp-content/uploads/2015/06/Lopa-Shah-ACEF-Presentation_Submitted.pdf	Opened; see research file
DE:S3	Ecofin Agency, "Zambia's long-delayed Ngonye hydropower project secures new investor", 19 June 2026. https://www.ecofinagency.com/news-infrastructures/1906-56617-zambias-long-delayed-ngonye-hydropower-project-secures-new-investor	Opened; see research file
DE:S4	Business Front, "Uganda commissions \$20 million Nyagak III hydropower project". https://businessfront.com/energy/news/uganda-commissions-20-million-nyagak-iii-hydropower-project-to-boost-electricity-supply/	Opened; see research file
DE:S5	WikiProcedure, "Uganda: Apply for Electricity Generation License". https://wikiprocedure.com/index.php/Uganda_-_Apply_for_Electricity_Generation_License	Opened; see research file
DE:S8	ALN, <i>Uganda Power Guide 2024</i> . https://aln.africa/wp-content/uploads/2024/06/ALN-Uganda-Power-Guide-2024.pdf	Opened; see research file
DE:S9	ERA / KfW, <i>GET FiT Program Uganda Overview Brief</i> . https://www.era.go.ug/wp-content/uploads/2025/05/GET-FiT-Info-Brief.pdf	Opened; see research file
DE:S11	Wikipedia, Bugoye Hydroelectric Power Station. https://en.wikipedia.org/wiki/Bugoye_Hydroelectric_Power_Station	Search summary only
DE:S14	IED, Small Hydro Power Projects for tea factories, Kenya and Uganda (2007). https://extranet.ied-sa.fr/en/projects-and-references/references/details/10/85/fiches-gb-2007.html	Opened; see research file
DE:S15	Hydropower & Dams, Sinoe Rapids small hydro (Liberia). https://www.hydropower-dams.com/?p=1009	Opened; see research file
DE:S16	Water Power Magazine, Kakono hydropower FS/ESIA (Tanzania). https://www.waterpowermagazine.com/?p=10699	Opened; see research file
DE:S17	AllAfrica / AfDB, "SEFA grants US\$1 million to prepare an innovative community-owned hydropower project in Kenya", Jan 2017. https://allafrica.com/stories/201701060627.html	Opened; see research file
DE:S18	Engineering News, "Financing and legal frameworks key for successful renewables projects in Africa", 19 July 2016 (Castalia). https://www.engineeringnews.co.za/article/financing-and-legal-frameworks-key-for-successful-renewables-projects-in-africa-2016-07-19	Opened; see research file
DE:S21	Cities Climate Finance Leadership Alliance, Africa50 Project Development. https://citiesclimatefinance.org/project-preparation-resource-directory/africa50-project-development	Opened; see research file
DE:S23	Brookings, "Figures of the week: Africa's infrastructure paradox", 24 Feb 2021 (citing McKinsey 2020). https://www.brookings.edu/articles/figures-of-the-week-africas-infrastructure-paradox	Opened; see research file

ID	Source	Verification
DE:S30	FMO project detail, Bugoye Hydro Limited (51836). https://www.fmo.nl/project-detail/51836	Opened; see research file
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Annex M. Glossary

Term	Meaning
Availability or capacity payment	Payment for a plant being available, whether or not energy is taken
CFADS	Cash flow available for debt service
Commercial operation date (COD)	The date the plant starts selling energy under the PPA
Conditions precedent	Conditions that must be met before lenders are obliged to fund
Co-developer	An investor who funds part of development in exchange for a share of the project
Deemed energy	Energy paid for but not taken, because of grid or buyer failure
Development premium	A payment to the developer at close, above the reimbursement of development costs
DFI	Development finance institution
DSCR	Debt service coverage ratio
DSRA	Debt service reserve account
EPC	Engineering, procurement and construction contract
ESIA	Environmental and social impact assessment
Financial close	The point at which all conditions precedent are met and lenders are obliged to fund
Gearing	Debt as a share of total funding
IDC	Interest during construction
IPP	Independent power producer
LLCR	Loan life coverage ratio
Locked financing	Stress testing with the financing package held at its base-case terms
P50, P90	Generation exceeded with 50 and 90 percent probability
PPA	Power purchase agreement
PPP	Public-private partnership
Reference class	A set of comparable past projects used to estimate overruns or delays
Risk-weighted value	A value that weights each outcome by its probability
Run-of-river	A plant with little or no storage, generating with the river's flow
Sculpting	Setting debt service in each period as CFADS divided by a target DSCR
Sell-down	Sale of part of the developer's stake, usually after commercial operation
Step-up	The rise in value when a risk falls away, at close or at commercial operation